

HASHEM A. DAMRAH

University of Oregon
Department of Physics
Eugene, OR 97403, USA

Abstract

Gravitational waves, ripples in the fabric of spacetime caused by accelerating masses, have become a cornerstone of modern astrophysics and cosmology. This paper provides a comprehensive overview of the mathematical framework underlying gravitational waves, starting from the Einstein field equations and progressing through linearized gravity to the generation and detection of gravitational waves. We explore the theoretical background necessary to understand these phenomena, including the properties of spacetime curvature and the behavior of perturbations in the metric tensor. Additionally, we discuss recent developments in gravitational wave astronomy, highlighting key discoveries and their implications for our understanding of the universe. The mathematical perspective presented in this paper aims to deepen our understanding of gravitational waves and their significance in the broader context of physics and astronomy.

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CHAPTER 1

Preliminaries and Theoretical Background

1. Introduction

This paper is organized in two parts: chapter 1 goes over the preliminaries and theoretical background of gravitational waves, while chapter 2 delves into the Einstein field equations, linearized gravity, and recent developments in gravitational wave astronomy. We don't assume any prior knowledge (except the fundamentals, like calculus, linear algebra, etc), as we develop those throughout the course of the paper.

If you already have a strong understanding of differential geometry, you can skip to section 5, where we introduce the necessary physics background for understanding gravitational waves.

Please note that this paper will be more mathematical in nature, as I aim to explore the mathematical framework underlying gravitational waves, rather than focusing solely on their physical implications. The mathematical perspective will allow us to gain a deeper understanding of the properties and behavior of gravitational waves, as well as their significance in the broader context of physics and astronomy.

The goal is to provide a comprehensive overview of the mathematical framework underlying gravitational waves, while also highlighting their significance in the broader context of physics and astronomy. We start this chapter by going over the notation used throughout this paper.

2. Notation

Throughout this paper we employ the Einstein summation convention. Tensor components will be written using indices, where repeated indices appearing once as an upper index and once as a lower index imply summation over their range.

2.1. Basic Index Notation. More precisely, if an index μ appears exactly twice in a term, once as a superscript and once as a subscript, then:

$$A^\mu B_\mu := \sum_{\mu=0}^{n-1} A^\mu B_\mu.$$

Unless otherwise stated, Greek indices μ, ν, ρ, σ range from 0 to $n - 1$, while Latin indices i, j, k range from 1 to $n - 1$. We adopt the convention that repeated indices are summed only when one index is upper and the other lower; repeated indices both upper or both lower do not imply summation.

2.2. The Kronecker Delta and Identity. The *Kronecker delta*, denoted by δ^μ_ν , serves as the identity operator for index substitution:

$$\delta^\mu_\nu = \begin{cases} 1 & \text{if } \mu = \nu \\ 0 & \text{if } \mu \neq \nu \end{cases}.$$

In our index notation, the delta satisfies the property $A^\mu \delta_\mu^\nu = A^\nu$. This is particularly useful when simplifying coordinate transformations or defining the Euclidean metric δ_{ij} in flat background regions of a medical scan.

2.3. Raising and Lowering Indices. Indices are raised and lowered using the *metric tensor* $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$. The position of indices is significant, as it distinguishes between covariant (downstairs) and contravariant (upstairs) components of tensors:

$$V_\mu = g_{\mu\nu} V^\nu, \quad \omega^\mu = g^{\mu\nu} \omega_\nu.$$

We require the metric and its inverse to satisfy $g_{\mu\rho} g^{\rho\nu} = \delta_\mu^\nu$.

2.4. Partial Derivatives and Shorthand Notation. Partial derivatives with respect to coordinates x^μ will be written in the standard form $\frac{\partial f}{\partial x^\mu}$. For notational convenience, we introduce the shorthand:

$$\partial_\mu f := \frac{\partial f}{\partial x^\mu} \quad \text{and} \quad \partial_{y^\mu} f := \frac{\partial f}{\partial y^\mu},$$

when differentiation with respect to a different coordinate system is required. We also use the shorthand notation $\dot{x}$ to denote differentiation with respect to the parameter of a curve (typically λ , τ , or t):

$$\dot{x} := \frac{dx}{d\lambda}.$$

Example 1.1. *The Euler-Lagrange equations for a Lagrangian $L(x^k, \dot{x}^k)$ are:*

$$\frac{\partial L}{\partial x^k} - \frac{\partial}{\partial \lambda} \left(\frac{\partial L}{\partial \dot{x}^k} \right) = 0.$$

Using our shorthand notation, this is compactly written as:

$$\partial_k L - \frac{d}{d\lambda} (\partial_{\dot{x}^k} L) = 0.$$

2.5. Symmetrization and Anti-symmetrization. To describe geometric objects such as the second fundamental form or the curvature tensor, we use parentheses and brackets to denote the symmetric and anti-symmetric parts of a tensor:

$$T_{(\mu\nu)} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu}), \quad T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}).$$

2.6. More In-depth Example: The Scalar Product. The Riemannian inner product of two vectors U, V (which might represent the initial velocity of an ultrasound pulse and the normal to the skin) is written as:

$$\langle U, V \rangle_g = g_{\mu\nu} U^\mu V^\nu.$$

In an isotropic Euclidean medium where $g_{\mu\nu} = \delta_{\mu\nu}$, this reduces to the standard dot product $U^\mu V_\mu$. The magnitude of the gradient of a pressure field f is given by:

$$|\nabla f|^2 = g^{\mu\nu} (\partial_\mu f) (\partial_\nu f).$$

This demonstrates how the inverse metric $g^{\mu\nu}$ is necessary to compute the norm of a covector (gradient).

3. Differential Geometry

We start this section by developing the necessary mathematical tools and concepts required to understand gravitational waves. This section just briefly introduces concepts from differential geometry that are used throughout this paper.

3.1. Intro to Manifolds. Let's begin by introducing the concept of a manifold.

Definition 1.2. An n -dimension manifold, M , is a topological space such that

- (1) M is locally Euclidean, i.e., for every point $p \in M$, there exists an open neighborhood U of p and a homeomorphism $\varphi : U \rightarrow V \subset \mathbb{R}^n$, where V is an open subset of $\mathbb{R}^n$. We call the pair (U, φ) a chart of M .
- (2) M is Hausdorff, i.e., for any two distinct points $p, q \in M$, there exist disjoint open neighborhoods U_p and U_q such that $p \in U_p$ and $q \in U_q$.
- (3) M is second countable, i.e., there exists a countable basis for the topology of M .

We define an *atlas* as the collection of all the charts along with their transition maps, $\mathcal{A} = \{(U_i, \varphi_i)\}$. To go from one chart, U_i , to another chart, U_j , if $U_i \cap U_j \neq \emptyset$, we can use the transition maps:

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j).$$

But, throughout this paper, we only care about smooth manifolds, which are manifolds that are endowed with a smooth structure. Before doing so, we first need to define what a smooth map is.

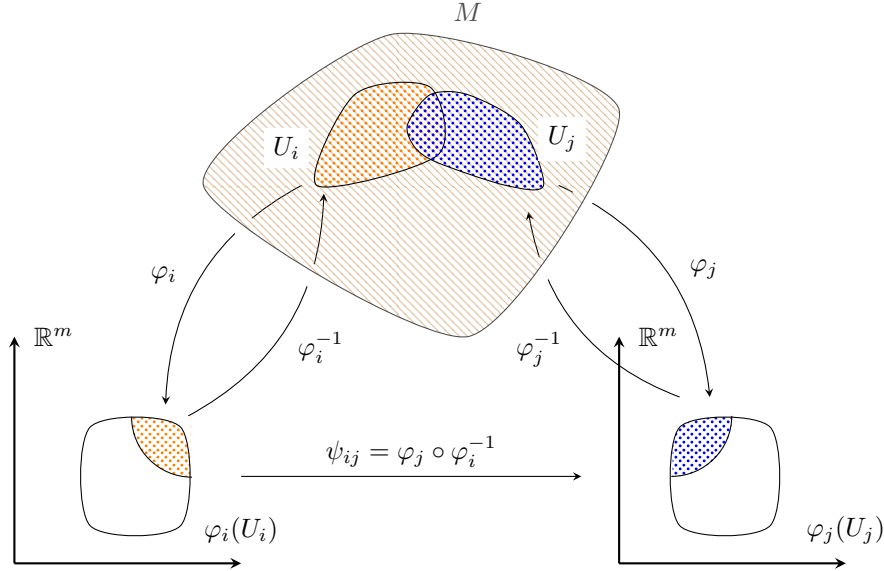


FIGURE 1. Transition maps between charts on a manifold.

Definition 1.3. If U and V are open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, then $F : U \rightarrow V$ is said to be smooth¹ if each of F 's component functions are continuous and have

¹Some authors use "smooth" to denote infinitely differentiable (C^∞), some use it to indicate just k -differentiable (C^k). I use it to denote C^∞ .

continuous partial derivatives. In addition, if F is bijective, then F is called a diffeomorphism.

We can formally define a smooth manifold as follows.

Definition 1.4. A k -differentiable manifold is a manifold M equipped with an atlas $\mathcal{A}$ such that for every non-disjoint pair of charts (U_i, φ_i) and (U_j, φ_j) in $\mathcal{A}$, the transition map $\varphi_j \circ \varphi_i^{-1}$ is a k -differentiable function. If $k = \infty$, we call M a smooth manifold.

This section's all about defining what it means to have a smooth function from a manifold M to a Euclidean space E along with a map from a manifold M to another manifold N .

Definition 1.5. Suppose M is a manifold². Let $f : M \rightarrow \mathbb{R}^k$, for some $k \geq 1$. We say that f is a smooth function if, for every $p \in M$, there exists a smooth chart (U, φ) , where $p \in U$, such that $f \circ \varphi^{-1}$ is smooth on the open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

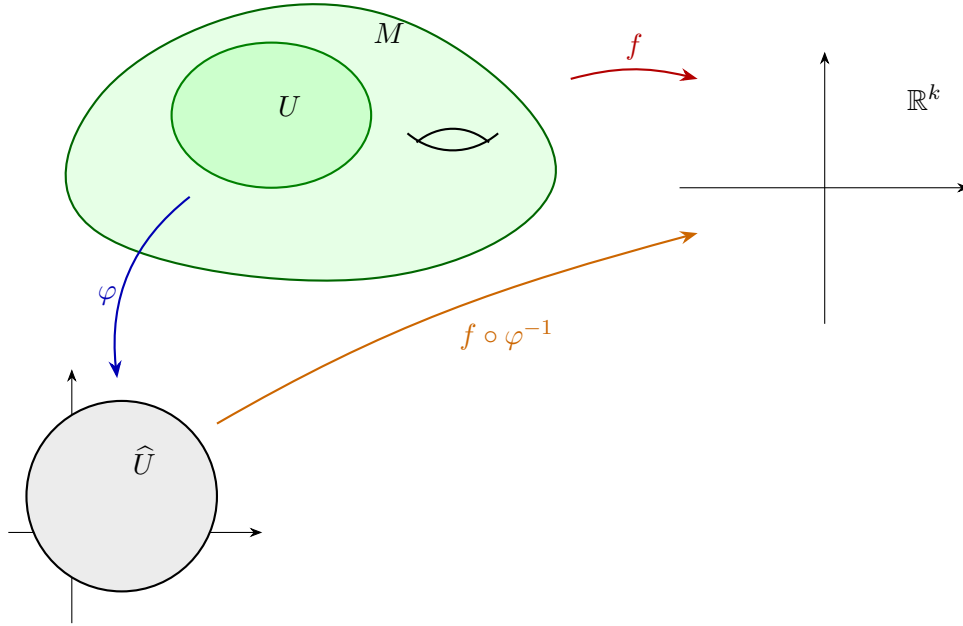


FIGURE 2. Smooth functions

Now, we talk about smooth maps between manifolds.

Definition 1.6. Let M and N be two different manifolds. We say $F : M \rightarrow N$ is a smooth map if for every $p \in M$, there exists charts (U, φ) , where $p \in U$, and (V, ψ) , where $F(p) \in V$ such that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

A nice property (which we won't be proving), is that a composition of smooth maps is also smooth.

²Throughout this paper, we're only interested in smooth manifolds, so we will assume that all manifolds are smooth unless otherwise stated. We also drop the adjective " n -dimension". So, it's safe to assume that whenever we say "manifold", we mean a smooth manifold of dimension n , unless otherwise stated.

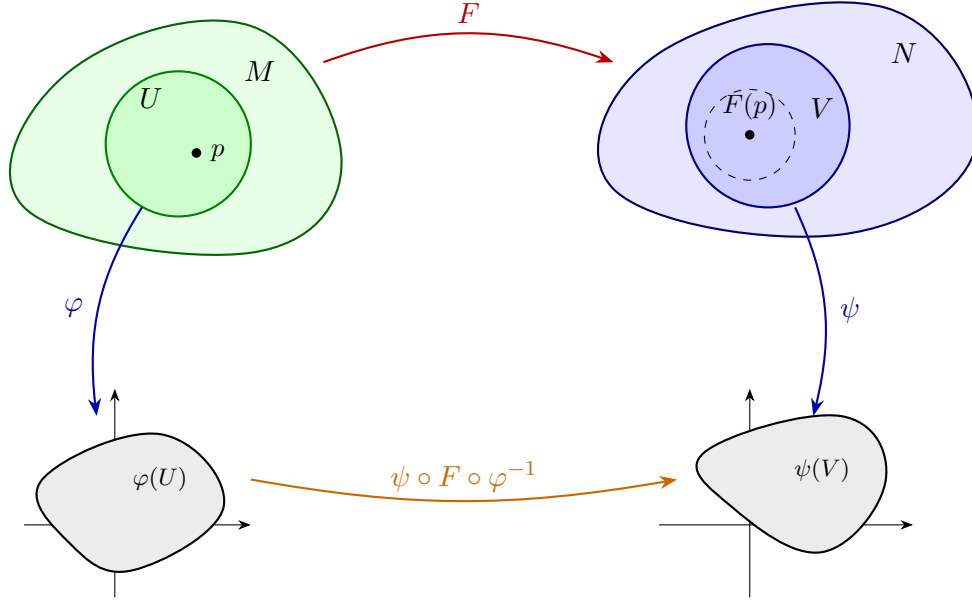
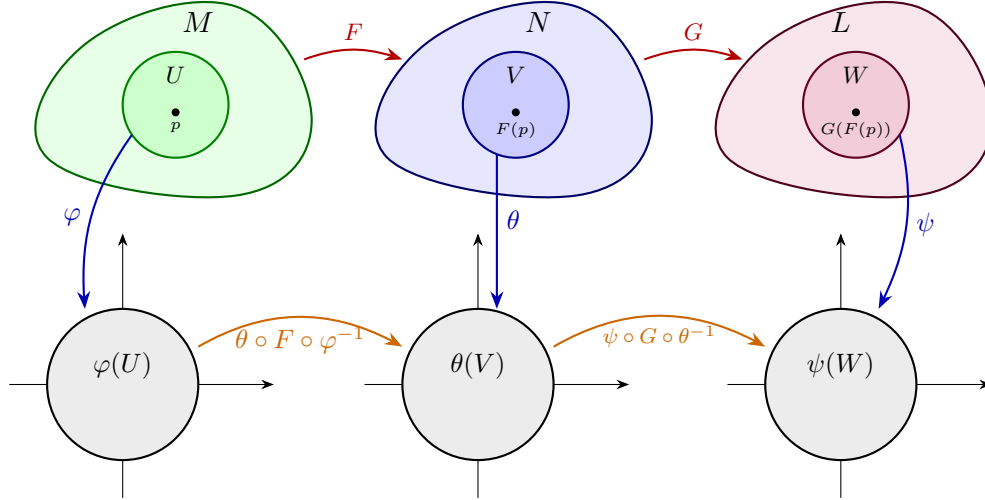
FIGURE 3. Smooth maps between M and N 

FIGURE 4. Composition of smooth maps

3.2. Tangent Bundles, Cotangent Bundles, and Sections. Our next main definition is that of a tensor, but in order to do so, we need to introduce a few more concepts, such as the tangent and cotangent bundles of a manifold.

Definition 1.7. Let M be a manifold and $p \in M$. The tangent space at p , denoted $T_p M$, is the vector space of derivations at p , that is,

$$T_p M = \{X : C^\infty(M) \rightarrow \mathbb{R} \mid X \text{ is linear and satisfies the Leibniz rule}\}.$$

That is, for all $f, g \in C^\infty(M)$,

$$X(fg) = X(f)g(p) + f(p)X(g).$$

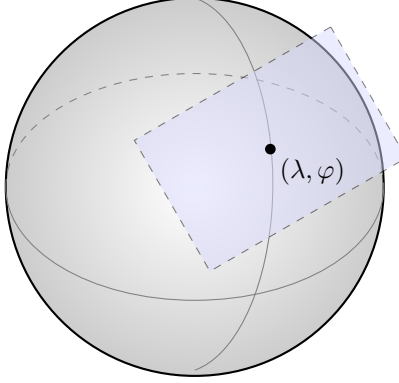


FIGURE 5. Tangent space of $\mathbb{S}^2$

A rather straightforward basis for the tangent space is given by

$$\mathcal{B} = \left\{ \frac{\partial}{\partial x^i} \Big|_a \right\}_{i=1}^{i=n} = \{\partial_i(a)\}_{i=1}^{i=n},$$

defined by

$$\frac{\partial}{\partial x^i} \Big|_a f = \frac{\partial f}{\partial x^i}(a) = \partial_i(a),$$

using Einstein notation. This is called the *coordinate basis* for $T_p M$. We can also define the tangent space using equivalence classes of curves, but we won't be doing that here.

Our next definition will be the tangent bundle. But before we do that, we need to define what a disjoint union is.

Definition 1.8. The disjoint union of a family of sets $\{A_i\}_{i \in I}$ is the set

$$\bigsqcup_{i \in I} A_i = \{(i, a) : i \in I, a \in A_i\}.$$

Example 1.9. If we have two sets $A = \{1, 2\}$ and $B = \{1, 2\}$, then their disjoint union is

$$A \sqcup B = \{(A, 1), (A, 2), (B, 1), (B, 2)\}.$$

Notice that the elements $A = B$, but the disjoint union keeps track of which set each element belongs to. If we do $(A, 1) \cap (B, 1) = \emptyset$, even tho 1 does indeed equal 1.

To study velocities and gradients across the entire manifold, we must group all individual tangent spaces into a single geometric object. We begin with the Tangent Bundle, which serves as our primary example of a more general structure known as a Vector Bundle.

Definition 1.10. A tangent bundle, denoted as TM , of a manifold M is defined as

$$TM = \bigsqcup_{p \in M} T_p M = \{(p, v) : p \in M, v \in T_p M\}.$$

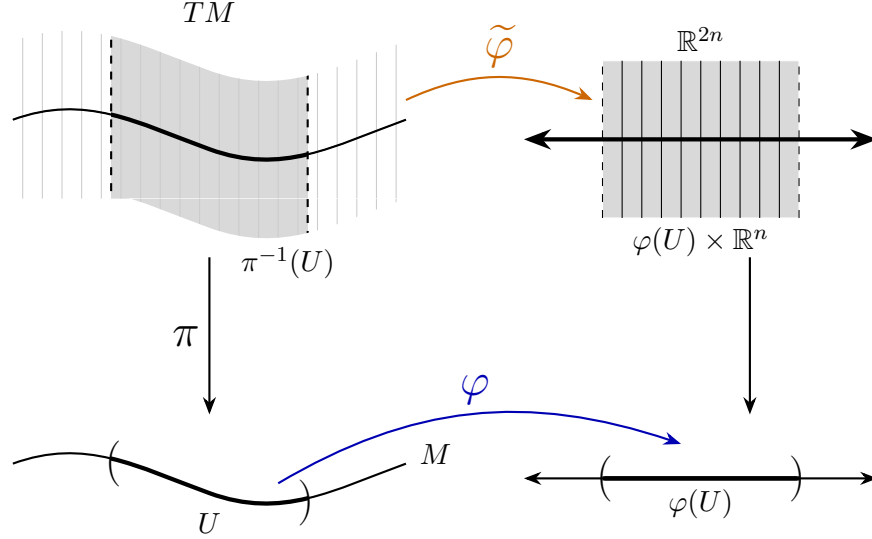


FIGURE 6. Coordinates for the Tangent bundle

The tangent bundle is the prototypical example of a vector bundle, a structure that allows us to attach a vector space to every point of a manifold in a way that varies smoothly.

Definition 1.11. A vector bundle of rank k over M is a manifold E (the total space) and a smooth surjection $\pi : E \rightarrow M$ (the projection) such that:

- (1) For each $p \in M$, the set $E_p = \pi^{-1}(p)$ is a k -dimensional vector space, called the fiber over p .
- (2) E is locally trivial: every $p \in M$ has a neighborhood U and a diffeomorphism $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ that is a linear isomorphism on each fiber.

NOTE 1. In this framework, the tangent bundle TM is a vector bundle where the fibers are the tangent spaces $T_p M$. Similarly, we define the cotangent bundle T^*M as the bundle whose fibers are the dual spaces $T_p^* M$, housing the covectors (gradients).

Definition 1.12. The cotangent space $T_p^* M$ is the dual space to $T_p M$. If $\{\partial_i\}$ is the coordinate basis for the tangent space, we define the coordinate dual basis $\{dx^i\}$ for $T_p^* M$ such that:

$$dx^i(\partial_j) = \delta_j^i,$$

where δ_j^i is the Kronecker delta. This allows us to express any covector $\omega \in T_p^* M$ as $\omega = \omega_i dx^i$.

NOTE 2. If $\{\partial_i\}$ is a basis for $T_p M$, then the dual basis $\{dx^i\}$ is a basis for $T_p^* M$.

NOTE 3. Covectors live in the cotangent bundle, while the contravariant vectors (i.e., tangent vectors) live in the tangent bundle. The contravariant vectors represent velocities or directions. The covariant represents gradients or linear functionals.

In local coordinates, tensor components are written using upper and lower indices to distinguish components with respect to a chosen basis of $T_p M$ and its dual basis in $T_p^* M$.

Definition 1.13. Let $\pi : E \rightarrow M$ be a vector bundle over M . A section of E is a smooth map

$$s : M \rightarrow E,$$

such that

$$\pi \circ s = \text{id}_M.$$

The space of smooth sections of E is denoted $\Gamma(E)$.

3.3. Tensors. Now, with all that out of the way, we can finally define a tensor.

Definition 1.14. Let M be a manifold and $p \in M$. An (r, s) -tensor at p is a multilinear map

$$T_p : (T_p^* M)^r \times (T_p M)^s \rightarrow \mathbb{R}.$$

Definition 1.15. The tensor bundle of type (r, s) over a manifold M is the vector bundle

$$T_s^r M = \bigsqcup_{p \in M} T_s^r(p) = \bigsqcup_{p \in M} \left(\underbrace{T_p^* M \otimes \cdots \otimes T_p^* M}_{r \text{ times}} \otimes \underbrace{T_p M \otimes \cdots \otimes T_p M}_{s \text{ times}} \right).$$

NOTE 4. We use the shorthand notation

$$A^{\otimes r} = \underbrace{A \otimes \cdots \otimes A}_{r \text{ times}}.$$

Thus the tensor bundle may be written as

$$T_s^r M = \bigsqcup_{p \in M} (T_p^* M)^{\otimes r} \otimes (T_p M)^{\otimes s}.$$

Definition 1.16. An (r, s) -tensor field on M is a smooth section of the tensor bundle

$$\pi : T_s^r M \rightarrow M.$$

That is, a smooth map

$$T : M \rightarrow T_s^r M,$$

such that

$$\pi \circ T = \text{Id}_M.$$

In local coordinates $\{x^i\}$, an (r, s) -tensor field T is expressed as a linear combination of basis tensor products:

$$T = T_{j_1 \dots j_s}^{i_1 \dots i_r} (\partial_{i_k})^{\otimes r} \otimes (dx^{j_m})^{\otimes s}.$$

Here, the functions $T_{j_1 \dots j_s}^{i_1 \dots i_r}$ are the *components* of the tensor. Under a change of coordinates $x \rightarrow \tilde{x}$, these components transform according to the rule:

$$\tilde{T}_{b_1 \dots b_s}^{a_1 \dots a_r} = T_{j_1 \dots j_s}^{i_1 \dots i_r} \frac{\partial \tilde{x}^{a_1}}{\partial x^{i_1}} \cdots \frac{\partial \tilde{x}^{a_r}}{\partial x^{i_r}} \frac{\partial \tilde{x}^{j_s}}{\partial x^{b_s}}.$$

Here are a few examples of tensors:

Example 1.17.

- **The Metric Perturbation Tensor ($h_{\mu\nu}$):** This is a symmetric, rank-2 tensor that represents the “ripples” in spacetime. In the weak-field limit, the full metric is decomposed as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. In the Transverse-Traceless (TT) gauge, for a wave propagating in the z -direction, it reduces to:

$$h_{ij}^{TT} = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This tensor directly encodes the two physical polarization states, h_+ and $h_\times$, which describe the stretching and squeezing of space.

- **The Energy-Momentum Tensor ($T_{\mu\nu}$):** This rank-2 tensor serves as the “source” for gravitational waves. It encapsulates the density and flux of energy and momentum in spacetime. While General Relativity primarily utilizes its symmetric part $T_{(\mu\nu)}$, the **GQFT** framework incorporates the antisymmetric part $T_{[\mu\nu]}$ to account for the intrinsic spin-currents of particles, which act as sources for vector-type gravitational radiation.
- **The Riemann Curvature Tensor ($R_{\sigma\mu\nu}^\rho$):** A rank-4 tensor that provides a complete description of the local curvature of the manifold. In the context of detection, it is used in the Geodesic Deviation Equation:

$$\frac{D^2 S^\mu}{d\tau^2} = R_{\nu\rho\sigma}^\mu U^\nu U^\rho S^\sigma.$$

This tensor is the bridge between the abstract metric perturbation and the physical tidal forces that move the mirrors in an interferometer like LIGO.

To manipulate the physical quantities represented by these tensors, we define several fundamental operations:

- **Einstein Summation Convention:** Throughout this paper, we adopt the convention that whenever an index appears twice in a single term – once as a superscript and once as a subscript – a summation over the full range of that index is implied. For example, $X^i \omega_i$ denotes $\sum_{i=1}^n X^i \omega_i$ (cf Section 2 for more details).
- **Contraction:** A contraction is an operation that reduces the rank of an (r, s) -tensor to $(r - 1, s - 1)$. Locally, this is done by setting one upper index equal to one lower index and summing. For a $(1, 1)$ -tensor T_j^i , the contraction is the scalar T_i^i , analogous to the trace of a matrix.
- **Tensor Product:** Given an (r, s) -tensor T and a (p, q) -tensor U , their tensor product $T \otimes U$ is an $(r + p, s + q)$ -tensor. This allows us to construct complex physical models, such as the metric tensor, from simpler 1-forms.

NOTE 5. Once a Riemannian metric g is introduced in Section 4, we gain the ability to raise and lower indices. This operation, sometimes called the musical isomorphisms ($\flat$ and $\sharp$), allows us to physically identify vectors with covectors, effectively using the internal geometry of the tissue to translate between directional velocities and gradients of potential.

3.4. Differential Forms and the Wedge Product. Next, we introduce the concept of a differential form. A very well-known example of a differential form is dx , which is a 1-form. Formally, we can define a 1-form as follows.

Definition 1.18. A 1-form on a manifold M is a smooth section of the cotangent bundle T^*M , that is,

$$\omega \in \Gamma(T^*M),$$

where $\omega : M \rightarrow T^*M$ is a smooth map such that $\pi \circ \omega = \text{id}_M$. Equivalently, for each $p \in M$, a 1-form assigns a covector $\omega_p \in T_p^*M$ smoothly in p .

A 1-form represents a linear functional that takes a tangent vector at a point and produces scalar.

Example 1.19. In local coordinates $\{x^\mu\}$, a 1-form ω can be written as

$$\omega = \omega_\mu dx^\mu.$$

If $v = v^\mu \partial_\mu \in T_pM$, then

$$\omega(v) = \omega_\mu v^\mu.$$

We denote the space of smooth 1-forms on M by $\Omega^k(M)$, with $k = 1$. This implies that there are higher order forms. So, we need a way to generalize this to higher-order forms, which is where the concept of the wedge product comes in. The wedge product allows us to combine 1-forms to create higher-order forms, such as 2-forms, 3-forms, etc. We can define the wedge product as follows.

Definition 1.20. The wedge product

$$\wedge : \Omega^p(M) \times \Omega^q(M) \rightarrow \Omega^{p+q}(M),$$

is the bilinear operation that takes a p -form α and a q -form β and produces a $(p+q)$ -form $\alpha \wedge \beta$, defined by the following properties:

- (1) Bilinearity: For all $\alpha, \alpha' \in \Omega^p(M)$, $\beta, \beta' \in \Omega^q(M)$, and $f, g \in C^\infty(M)$,

$$(f\alpha + g\alpha') \wedge \beta = f(\alpha \wedge \beta) + g(\alpha' \wedge \beta),$$

$$\alpha \wedge (f\beta + g\beta') = f(\alpha \wedge \beta) + g(\alpha \wedge \beta').$$

- (2) Antisymmetry: For all $\alpha \in \Omega^p(M)$ and $\beta \in \Omega^q(M)$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

In particular, if α is a 1-form, then $\alpha \wedge \alpha = 0$.

- (3) Associativity: For all $\alpha \in \Omega^p(M)$, $\beta \in \Omega^q(M)$, and $\gamma \in \Omega^r(M)$,

$$(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma).$$

Now, we can use the wedge product to define a 2-form, which represents an orientated area element on the manifold. Just like in vector calculus, where we had $dA = dx dy$, we can define a 2-form ω as

$$\omega = \omega_{\mu\nu} dx^\mu \wedge dx^\nu.$$

Next, we need to define what the pullback of a differential form is and what a pushforward of a vector field is. The pullback allows us to take a differential form defined on one manifold and pull it back to another manifold via a smooth map, while the pushforward allows us to take a vector field defined on one manifold and push it forward to another manifold via a smooth map.

Definition 1.21. Let $f : M \rightarrow N$ be a smooth map between manifolds M and N . The pullback of a differential form ω on N by the map f is a differential form on M , denoted by $f^*\omega$, defined as follows: for any point $p \in M$ and any tangent vectors $v_1, v_2, \dots, v_k \in T_pM$, we have

$$(f^*\omega)_p(v_1, v_2, \dots, v_k) = \omega_{f(p)}(df_p(v_1), df_p(v_2), \dots, df_p(v_k)).$$

Example 1.22. Let $F : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ be defined by the polar coordinate transformation

$$F(r, \theta) = (x, y) = (r \cos \theta, r \sin \theta).$$

We compute the pullback of the coordinate 1-forms dx and dy .

Since

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta,$$

we have

$$dx = \cos \theta \, dr - r \sin \theta \, d\theta \quad \text{and} \quad dy = \sin \theta \, dr + r \cos \theta \, d\theta.$$

Therefore,

$$F^*(dx) = \cos \theta \, dr - r \sin \theta \, d\theta \quad \text{and} \quad F^*(dy) = \sin \theta \, dr + r \cos \theta \, d\theta.$$

As an application, consider the standard area form $dx \wedge dy$. Using the bilinearity and antisymmetry of the wedge product, we compute

$$\begin{aligned} F^*(dx \wedge dy) &= F^*(dx) \wedge F^*(dy) \\ &= (\cos \theta \, dr - r \sin \theta \, d\theta) \wedge (\sin \theta \, dr + r \cos \theta \, d\theta) \\ &= r \, dr \wedge d\theta. \end{aligned}$$

Thus, the pullback naturally produces the Jacobian factor r appearing in polar coordinates.

When we're computing pullbacks of differential forms, we can use the formula

$$f^* \left(\bigwedge_{i=1}^k du^i \right) = \bigwedge_{i=1}^k f^*(du^i) = \bigwedge_{i=1}^k \partial_j u^i \, dx^j.$$

Definition 1.23. Let $f : M \rightarrow N$ be a smooth map between manifolds M and N . The pushforward of a vector field X on M by the map f is a vector field on N , denoted by f_*X , defined as follows: for any point $p \in M$, we have

$$(f_*X)_{f(p)} = df_p(X_p).$$

Example 1.24. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$F(x, y) = (u, v) = (x^2 - y, x + y).$$

We compute the pushforward of the vector field $X = \partial_x$.

The Jacobian matrix of F is

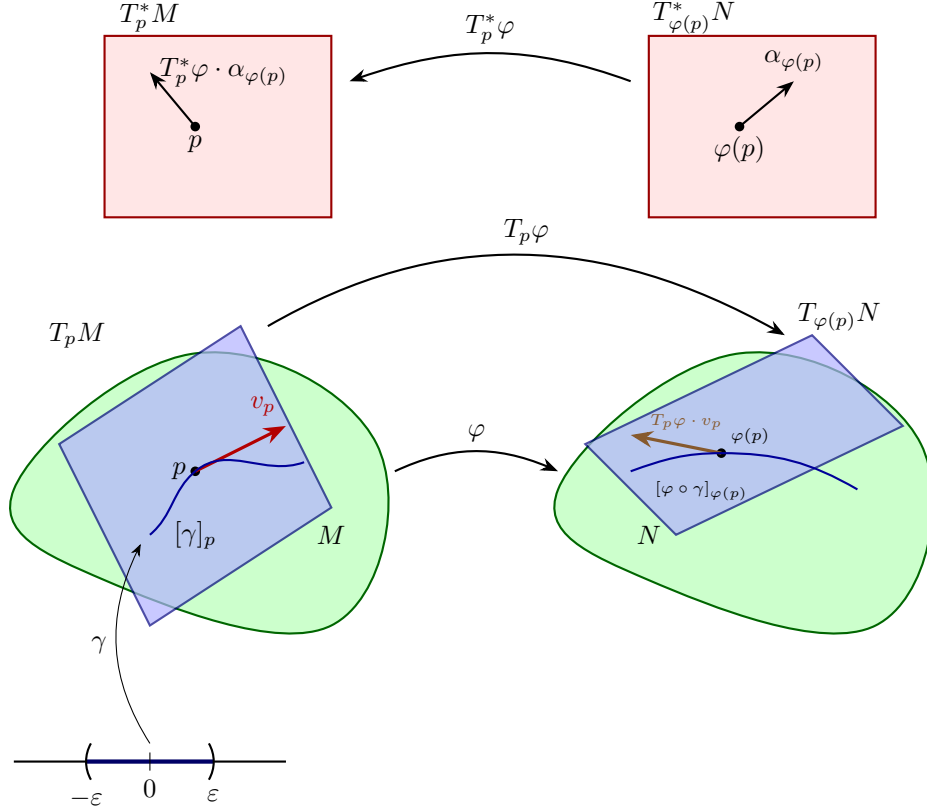
$$dF = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix} = \begin{pmatrix} 2x & -1 \\ 1 & 1 \end{pmatrix}.$$

Since ∂_x corresponds to the vector $(1, 0)$ in coordinates, we obtain

$$F_*(\partial_x) = 2x \, \partial_u + \partial_v.$$

Thus, the pushforward transforms the coordinate basis vector ∂_x into a linear combination of the coordinate basis vectors ∂_u and ∂_v .

NOTE 6. These pullbacks and pushforwards will be important later on when we talk about the behavior of gravitational waves under coordinate transformations. The pullback allows us to understand how differential forms, such as the metric tensor, transform under changes of coordinates, while the pushforward allows us to understand how vector fields, such as the velocity field of a particle, transform under changes of coordinates.

FIGURE 7. Pushforward and pullback under a smooth map $\varphi : M \rightarrow N$.

3.5. Differentials of Maps. We now define the differential of a map between two manifolds.

Definition 1.25. Let $F : M \rightarrow N$ be a smooth map between two manifolds M and N . For each point $p \in M$, we define the differential of F at p by

$$dF_p : T_pM \rightarrow T_{F(p)}N.$$

For any vector $v \in T_pM$, the image $v' = dF_p(v) \in T_{F(p)}N$ is defined by its action on smooth functions $g \in C^\infty(N)$:

$$(dF_p(v))(g) = v(g \circ F).$$

This definition ensures that the differential is a linear map that captures how the “directional derivative” at p translates to a directional derivative at $F(p)$.

In local coordinates $\{x^i\}$ on M and $\{y^\alpha\}$ on N , let F be represented by $y^\alpha = F^\alpha(x^1, \dots, x^n)$. If $v = v^i \partial_i$, then the components of the pushed-forward vector $v' = dF_p(v)$ are given by:

$$(v')^\alpha = \partial_i F^\alpha v^i.$$

Thus, the matrix of dF_p with respect to the coordinate bases is the Jacobian matrix $\partial_i F^\alpha$.

We finish this section by defining an immersion, which is a smooth map that locally looks like an embedding but may not be globally injective.

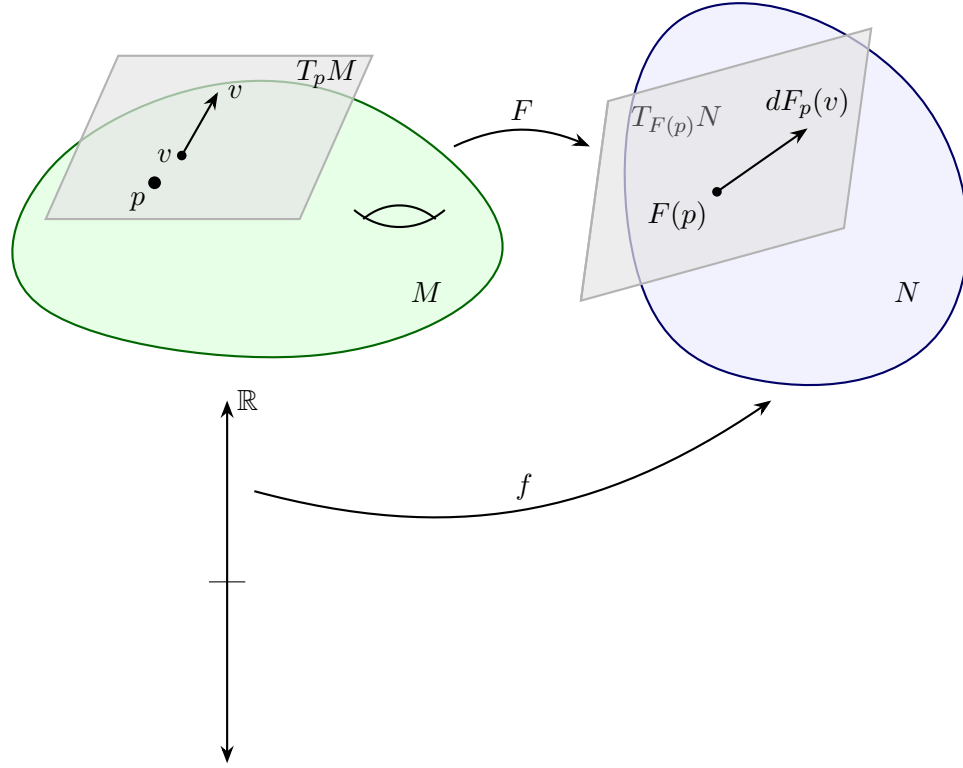


FIGURE 8. The differential

Definition 1.26. A smooth map $F : M \rightarrow N$ is called an *immersion* if its differential $dF_p : T_p M \rightarrow T_{F(p)} N$ is injective for all $p \in M$.

Definition 1.27. A smooth map $F : M \rightarrow N$ is called a *submersion* if its differential $dF_p : T_p M \rightarrow T_{F(p)} N$ is surjective for all $p \in M$.

Definition 1.28. A smooth embedding is an *injective immersion* $F : M \rightarrow N$ that is also a topological embedding (i.e., a homeomorphism onto its image with the subspace topology).

Definition 1.29. We say $S \subset N$ is an *embedded submanifold* if it is the image of a smooth embedding. If $F : M \rightarrow N$ is merely an injective immersion, we call its image an *immersed submanifold*.

One useful result we'll be using quite frequently in Chapter 2 is the following:

Corollary 1.30. Suppose $F : M \rightarrow N$ is a smooth map, $p \in M$, and $v \in T_p M$. Then

$$dF_p(v) = (f \circ \gamma)'(0),$$

for any smooth curve $\gamma : [0, 1] \rightarrow M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

3.6. Connections and the Riemann Tensor. To describe the geometry of a manifold, we must introduce a way to differentiate vector fields. Unlike in Euclidean space, there is no canonical way to compare vectors at different points without a “connection.”

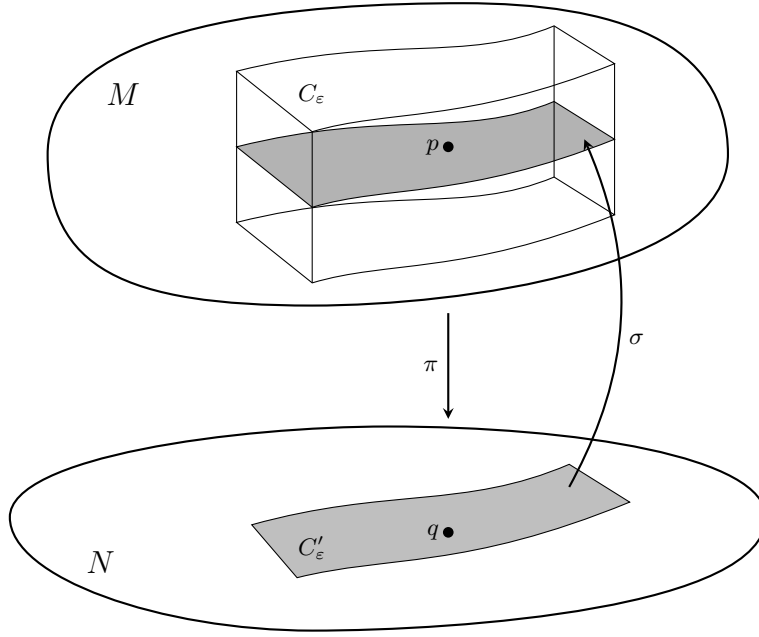


FIGURE 9. Local section of a submersion

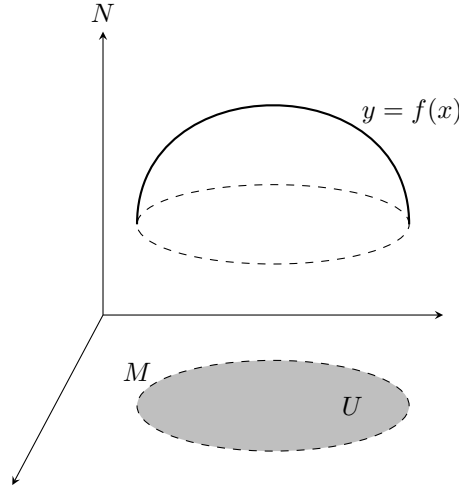


FIGURE 10. An embedded submanifold

Definition 1.31. A connection (or covariant derivative) on M is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM)$ satisfying linearity and the Leibniz rule:

$$\nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

The connection is $C^\infty(M)$ -linear in X , meaning $\nabla_{fX} Y = f\nabla_X Y$, but it satisfies the Leibniz rule in Y defined above. We need a way of measuring how much the connection fails to be symmetric. This is where the Lie bracket comes in, which measures the non-commutativity of vector fields.

Definition 1.32. The Lie bracket of two vector fields X and Y is the vector field defined by

$$[X, Y](f) = X(Y(f)) - Y(X(f)),$$

for all smooth functions $f \in C^\infty(M)$. The Lie bracket measures the non-commutativity of the flows generated by X and Y .

Geometrically, if we follow the flow of X for time t , then Y for time s , and then reverse the process, the Lie bracket $[X, Y]$ measures the displacement from the starting point.

We can also define the anti-symmetry of the connection using the Lie bracket. A connection is *symmetric* if $\nabla_X Y - \nabla_Y X = [X, Y]$ for all vector fields X and Y . If this condition is not satisfied, we can measure the failure of symmetry using the torsion tensor, which we will define shortly.

Definition 1.33. The Riemann curvature tensor associated to a connection ∇ is the $(1, 3)$ -tensor:

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

It measures the path-dependence of parallel transport and the failure of covariant derivatives to commute.

Here's a rough idea for what the parallel transport is (as we define it later on in Section 4): A vector field Z is *parallel* along a curve with tangent X if $\nabla_X Z = 0$. The Riemann tensor $R(X, Y)Z$ quantifies the change in Z when it is parallel transported around an infinitesimal loop.

Definition 1.34. The torsion tensor T is defined by $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$. A connection is torsion-free if $T = 0$.

While these definitions apply to any affine connection, physical applications in medical imaging—such as modeling signal propagation or tissue conductivity—require a specific connection that respects the internal “distance” of the manifold. In the following section, we introduce the Riemannian metric and the unique Levi-Civita connection associated with it.

3.7. Conformal and Isometric Maps. We conclude this section by defining conformal and isometric maps, which are the fundamental “equivalence relations” used to categorize Riemannian manifolds.

Definition 1.35. A smooth map $F : (M, g) \rightarrow (N, h)$ is called conformal if there exists a smooth function $\Omega : M \rightarrow (0, \infty)$, called the conformal factor, such that:

$$F^* h = \Omega^2 g.$$

In this case, F preserves the angles between any two tangent vectors but scales lengths by the factor $\Omega(p)$.

In medical imaging, we frequently work with *conformally flat* metrics. For example, in an isotropic ultrasound scan, we assume the manifold is a domain $M \subset \mathbb{R}^n$ with a metric $g = c(x)^{-2} \delta$, where $c(x)$ is the sound speed. This metric is conformal to the Euclidean metric δ ; thus, while the sound waves follow “straight-line” Euclidean angles locally, the “acoustic distance” (time-of-flight) is warped by the factor $1/c(x)$.

Definition 1.36. A smooth map $F : (M, g) \rightarrow (N, h)$ is called an isometry if:

$$F^*h = g.$$

This is a special case of a conformal map where $\Omega \equiv 1$. An isometry preserves both angles and lengths, making it a distance-preserving map: $d_h(F(p), F(q)) = d_g(p, q)$.

Example 1.37 (The Poincare Disk). A classic example of these concepts is the Poincare Disk model of hyperbolic space, (D, g_{hyp}) , where $g_{hyp} = \frac{4}{(1-|x|^2)^2} \delta$. This metric is conformal to the Euclidean disk. While the “Euclidean” straight lines represent one geometry, the “Hyperbolic” geodesics—which are circular arcs orthogonal to the boundary—represent a different metric that has been warped by a conformal factor that blows up at the boundary.

4. Riemannian Geometry

Now that we’ve covered the basics of differential geometry (cf Section 3), we now give a brief introduction to Riemannian Geometry.

While differential geometry gives us the structure of the manifold, *Riemannian geometry* introduces the tools necessary for measurement: lengths, angles, and volumes. Riemannian geometry coincides with our intuitive understanding that geometry is the study of lengths, angles, curvature, area, volume, and so on. It provides the mathematical framework for understanding the geometry of curved spaces, which is essential for many applications in physics, engineering, and other fields.

4.1. The Metric Tensor. Let M be a manifold. We can define something called a metric that allows us to talk about distances and angles on the manifold. We can define a metric as follows.

Definition 1.38. A metric tensor on M is a smooth assignment

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R},$$

for each point $p \in M$ such that:

- (1) g_p is bilinear,
- (2) g_p is symmetric,
- (3) g_p is nondegenerate.

Smoothness means that in any coordinate chart $\{x^\mu\}$, the components

$$g_{\mu\nu}(x) := g(\partial_\mu, \partial_\nu),$$

are smooth functions. From this, we get a special type of manifold, namely, the pseudo-Riemannian manifold.

Definition 1.39. The pair (M, g) is called a pseudo-Riemannian manifold.

If in addition g_p is positive definite for every $p \in M$, meaning

$$g_p(v, v) > 0 \quad \text{for all } v \neq 0,$$

then (M, g) is called a Riemannian manifold.

In general, a pseudo-Riemannian metric need not be positive definite. At each point p , the bilinear form g_p has a well-defined *signature* (p, q) , where p and q denote the number of positive and negative eigenvalues, respectively.

The metric allows us to:

- Measure lengths of tangent vectors:

$$\|v\|^2 = g(v, v),$$

- Measure angles between vectors,
- Raise and lower tensor indices using the inverse metric $g^{\mu\nu}$.

Because g is nondegenerate, there exists a unique inverse tensor $g^{\mu\nu}$ satisfying

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu.$$

This allows us to lower and raise indices:

$$v_\mu = g_{\mu\nu} v^\nu \quad \text{and} \quad v^\mu = g^{\mu\nu} v_\nu.$$

Metrics that are not positive definite naturally classify tangent vectors into different types depending on the sign of $g(v, v)$. A particularly important case is when the signature is $(1, n-1)$, which will be discussed in detail later when we introduce Lorentzian manifolds.

4.2. Levi-Civita Connection. Given a pseudo-Riemannian manifold (M, g) , there is a unique connection that is compatible with the metric and has zero torsion. This connection is called the *Levi-Civita connection*.

THEOREM 1.40 (Fundamental Theorem of Riemannian Geometry). *Let (M, g) be a pseudo-Riemannian manifold. There exists a unique connection ∇ (the Levi-Civita connection) such that:*

- (1) ∇ is torsion-free: $\nabla_X Y - \nabla_Y X = [X, Y]$.
- (2) ∇ is metric-compatible: $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

From the properties of the Levi-Civita connection, we can derive an explicit formula for ∇ in terms of the metric, known as the *Koszul formula*:

$$2g(\nabla_X Y, Z) = X(g(Y, Z)) + Y(g(Z, X)) - Z(g(X, Y)) + g([X, Y], Z) - g([Y, Z], X) - g([Z, X], Y).$$

In a local coordinate system $\{x^i\}$, we can express the covariant derivative of one basis vector ∂_j in the direction of another ∂_i . Since the result $\nabla_{\partial_i} \partial_j$ is itself a vector field, it can be written as a linear combination of the basis vectors $\{\partial_k\}$.

Definition 1.41. *The Christoffel symbols of the second kind are defined by the relation:*

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

Applying the Koszul formula to the coordinate basis vectors (where $[\partial_i, \partial_j] = 0$) yields the explicit formula:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}),$$

where g^{kl} is the inverse of the metric tensor g_{kl} . This formula allows us to compute the Christoffel symbols directly from the metric components and their derivatives.

Since the Levi-Civita connection is torsion-free, the Christoffel symbols are symmetric in the lower indices:

$$\Gamma_{ij}^k = \Gamma_{ji}^k.$$

This reduces the number of components you have to calculate (e.g., in 3D, there are only 18 independent components instead of 27).

Now, let's use the Christoffel symbols to compute the covariant derivative of a vector field $V = V^k \partial_k$ in the direction of another vector field $X = X^i \partial_i$. Let

$V = V^k \partial_k$ be a smooth vector field and $X = X^i \partial_i$ be the direction of differentiation. By the Leibniz rule (property of the connection), the covariant derivative $\nabla_X V$ is:

$$\nabla_X V = \nabla_{X^i \partial_i} (V^j \partial_j) = X^i \nabla_{\partial_i} (V^j \partial_j).$$

Applying the product rule:

$$\nabla_X V = X^i \left((\partial_i V^j) \partial_j + V^j \nabla_{\partial_i} \partial_j \right).$$

Recalling the definition of Christoffel symbols, $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$, we substitute this into the expression:

$$\nabla_X V = X^i \left((\partial_i V^k) \partial_k + V^j \Gamma_{ij}^k \partial_k \right).$$

Factoring out the basis ∂_k , we arrive at the component formula:

$$(\nabla_X V)^k = X^i \left(\frac{\partial V^k}{\partial x^i} + \Gamma_{ij}^k V^j \right).$$

4.3. Parallel Transport and Geodesics. In Euclidean space, a “straight” path is easily defined as one with zero acceleration. However, on a Riemannian manifold (M, g) , the notion of a straight line is complicated by the fact that the tangent spaces $T_p M$ and $T_q M$ at different points are distinct. To determine if a path is straight, we first need a way to move vectors from one point to another without “turning” them. This leads us to the concept of parallel transport.

Let (M, g) be a manifold equipped with the Levi-Civita connection ∇ . Given a smooth curve $\gamma : I \rightarrow M$ with tangent vector $\dot{\gamma}(t) = \frac{d\gamma}{dt}$, we seek to measure how a vector field X changes as we move along γ .

Definition 1.42. *Let X be a vector field defined along γ , such that $X(t) \in T_{\gamma(t)} M$. The covariant derivative of X along γ is the projection of the standard derivative back onto the tangent space:*

$$\frac{DX}{dt} := \nabla_{\dot{\gamma}(t)} X.$$

In local coordinates $\{x^\mu\}$, the covariant derivative accounts for the “twisting” of the coordinate basis through the Christoffel symbols $\Gamma_{\alpha\beta}^\mu$:

$$\left(\frac{DX}{dt} \right)^\mu = \frac{dX^\mu}{dt} + \Gamma_{\alpha\beta}^\mu \dot{\gamma}^\alpha X^\beta.$$

This formula captures the motivation for the connection: it corrects the ordinary derivative of the component X^μ by adding a term that represents how the basis vectors themselves are changing relative to the metric.

A vector is “parallel” if it does not change its orientation relative to the manifold’s geometry as it moves.

Definition 1.43. *A vector field X along γ is said to be parallel transported if its covariant derivative along the curve vanishes:*

$$\frac{DX}{dt} = 0 \quad \Leftrightarrow \quad \frac{dX^\mu}{dt} + \Gamma_{\alpha\beta}^\mu \dot{\gamma}^\alpha X^\beta = 0.$$

Because the Levi-Civita connection is metric-compatible, parallel transport is an *isometry* between tangent spaces. It preserves both the lengths of vectors and the angles between them:

$$\frac{d}{dt} g(X, Y) = g\left(\frac{DX}{dt}, Y\right) + g\left(X, \frac{DY}{dt}\right) = 0.$$

With parallel transport defined, we can now define the “straightest” possible curves on a manifold. A curve is a geodesic if it does not “accelerate” in any direction other than the one dictated by the curvature of the manifold itself.

Definition 1.44. A curve γ is a geodesic if its tangent vector is parallel transported along itself:

$$\frac{D\dot{\gamma}}{dt} = 0 \quad \text{or equivalently,} \quad \nabla_{\dot{\gamma}}\dot{\gamma} = 0.$$

Expanding this in local coordinates yields the *geodesic equation*, a second-order non-linear system of ODEs:

$$\frac{d^2x^\mu}{dt^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} = 0.$$

While the definition above is geometric (zero acceleration), geodesics also arise from a physical principle: the minimization of effort. Geodesics are critical points of the *energy functional*:

$$E[\gamma] = \frac{1}{2} \int_I g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu dt.$$

Applying the Euler-Lagrange equations to $E[\gamma]$ yields the exact same coordinate geodesic equation.

4.4. Curvature. Next, we introduce the concept of curvature on a manifold.

Definition 1.45. The Riemann curvature tensor is the $(1,3)$ -tensor $R : \Gamma(TM) \times \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TM)$ defined by:

$$(1.1) \quad R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

This tensor quantifies the failure of the second covariant derivatives to commute. Geometrically, it measures the change in a vector Z after it is parallel-transported around an infinitesimal loop spanned by X and Y .

In a local coordinate system $\{x^\mu\}$, the components of the Riemann tensor are computed from the Christoffel symbols and their partial derivatives:

$$R_{\sigma\mu\nu}^\rho = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda.$$

The Riemann tensor for a Levi-Civita connection satisfies several symmetries, most notably $R(X, Y)Z = -R(Y, X)Z$, which in coordinates reads $R_{\sigma\mu\nu}^\rho = -R_{\sigma\nu\mu}^\rho$. By contracting indices, we obtain the primary tools for geometric analysis:

- **Ricci Tensor:** $R_{\sigma\nu} = R_{\sigma\rho\nu}^\rho$. This $(0,2)$ -tensor represents the average curvature in a given direction.
- **Scalar Curvature:** $R = g^{\sigma\nu} R_{\sigma\nu}$. This is a single smooth function providing a global measure of how the manifold’s volume differs from Euclidean space.

Example 1.46. For a 2-sphere $\mathbb{S}^2$ of radius a with the standard metric $ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\varphi^2$, a direct computation of the Riemann components reveals that the curvature is constant. Specifically, the scalar curvature is:

$$R = \frac{2}{a^2}.$$

If a metric $g_{\mu\nu}$ is given, we may further contract to obtain the scalar curvature.

Definition 1.47. The scalar curvature is defined by

$$R = g^{\sigma\nu} R_{\sigma\nu}$$

While the Riemann tensor is a high-rank object, we can extract the curvature of a specific 2-dimensional plane $P \subset T_p M$ within the tangent space. This is often the most intuitive way to visualize curvature in higher dimensions.

Definition 1.48. Let $\{v, w\}$ be a basis for a 2-plane P in $T_p M$. The sectional curvature of P is defined by:

$$K(v, w) = \frac{g(R(v, w)w, v)}{g(v, v)g(w, w) - g(v, w)^2}.$$

For a Riemannian manifold, if K is constant for all planes at all points, the manifold is called a *space of constant curvature*. In imaging, sectional curvature helps characterize the “local shape” of an organ’s surface.

In dimensions $n \geq 4$, the Riemann tensor can be decomposed into a part determined by the Ricci tensor and a “remainder” known as the Weyl tensor.

Definition 1.49. The Weyl curvature tensor C is the traceless part of the Riemann tensor. It represents the component of curvature that distorts the shape of a volume element without changing its total volume.

This is particularly relevant for *conformal geometry*, where we study deformations that preserve angles but not necessarily distances—a common scenario when modeling tissue stretching.

Finally, for variational problems and physical modeling, we define a specific combination of the Ricci and Scalar curvatures.

Definition 1.50. The Einstein tensor G is the symmetric $(0, 2)$ -tensor defined by:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

The primary mathematical utility of G is that its divergence vanishes ($\nabla^\mu G_{\mu\nu} = 0$) due to the *contracted Bianchi identity*. This property is essential when setting up “conservation laws” for biological flows or deformations.

4.5. Curvature Contractions. The Riemann tensor possesses 256 components in four dimensions, but its inherent symmetries reduce this to only 20 independent components. These symmetries are essential for performing contractions correctly.

- **Algebraic Symmetries:**

$$(1.2) \quad R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} \quad (\text{Antisymmetry in first two indices})$$

$$(1.3) \quad R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu} \quad (\text{Antisymmetry in last two indices})$$

$$(1.4) \quad R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma} \quad (\text{Interchange symmetry})$$

- **The First Bianchi Identity:** This relates the cyclic permutation of the last three indices:

$$R_{\sigma\mu\nu}^\rho + R_{\mu\nu\sigma}^\rho + R_{\nu\sigma\mu}^\rho = 0$$

- **The Second (Differential) Bianchi Identity:** This describes how the curvature tensor varies across the manifold:

$$\nabla_\lambda R_{\sigma\mu\nu}^\rho + \nabla_\mu R_{\sigma\nu\lambda}^\rho + \nabla_\nu R_{\sigma\lambda\mu}^\rho = 0$$

By contracting the second Bianchi identity twice, we arrive at a result that is fundamental to variational principles in imaging (such as Total Variation denoising on manifolds):

$$(1.5) \quad \nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R.$$

This identity implies that the *Einstein tensor* $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$ is divergence-free ($\nabla^\mu G_{\mu\nu} = 0$). In physical modeling, this represents a local conservation law for the geometric “energy” of the manifold.

The Riemann tensor can be decomposed into three irreducible components that represent different geometric effects. This decomposition is particularly useful in higher-dimensional analysis (where $n \geq 3$).

- (1) **The Scalar Part:** Represents the portion of curvature associated with the scalar curvature R . This part determines how the volume of a small ball in the manifold compares to a ball of the same radius in Euclidean space.
- (2) **The Traceless Ricci Part:** Encapsulated by the tensor $R_{\mu\nu} - \frac{1}{n} R g_{\mu\nu}$. This describes the “tidal” forces or the anisotropic stretching of the volume that preserves the total volume but changes its shape.
- (3) **The Weyl Tensor ($C_{\rho\sigma\mu\nu}$):** Also known as the *conformal curvature tensor*. In dimensions $n \geq 4$, it is the part of the Riemann tensor that remains after the Ricci and Scalar parts are removed.

4.6. The Laplace-Beltrami Operator. In Euclidean space, the Laplacian $\Delta = \sum \partial_i^2$ is the most fundamental second-order differential operator. On a Riemannian manifold (M, g) , this is generalized to the *Laplace-Beltrami operator* Δ_g , which accounts for the varying metric and curvature of the space.

Definition 1.51. *The Laplace-Beltrami operator acting on a smooth function $f \in C^\infty(M)$ is defined as the divergence of the gradient:*

$$(1.6) \quad \Delta_g f = \operatorname{div}(\operatorname{grad} f) = \nabla^\mu \nabla_\mu f.$$

Using the metric g and its determinant $|g| = \det(g_{\mu\nu})$, the Laplace-Beltrami operator can be written explicitly without requiring the Christoffel symbols:

$$(1.7) \quad \Delta_g f = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu f \right).$$

This formula is particularly powerful for medical imaging simulations because it only requires the metric components $g_{\mu\nu}$ and their first derivatives.

The study of the equation $\Delta_g \varphi = -\lambda \varphi$ is known as the *spectral geometry* of the manifold. For a compact manifold, the eigenvalues $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \dots$ form a discrete spectrum that encodes global geometric information, such as the volume and total curvature (by the Heat Kernel expansion).

5. Theoretical Physics Background

Now that we have the mathematical preliminaries out of the way, we can start discussing the theoretical background of gravitational waves. This section will cover the fundamental concepts and equations that form the basis of our understanding of gravitational waves, in addition to a more thorough explanation to working with Einstein’s notation and the Einstein summation convention.

5.1. Review of Special Relativity and Minkowski Spacetime. The transition from Riemannian geometry to physical theory begins with Special Relativity (SR). In this framework, we model the universe as a four-dimensional manifold known as Minkowski Spacetime $(\mathcal{M}, \eta)$. Unlike the Riemannian manifolds discussed in Section 4, spacetime is pseudo-Riemannian with a signature of $(1, 3)$ or $(-, +, +, +)$.

In a standard inertial coordinate system $\{x^0, x^1, x^2, x^3\} = \{ct, x, y, z\}$, the components of the flat-space metric $\eta_{\mu\nu}$ are given by the constant matrix:

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This metric defines the spacetime interval ds^2 between two events:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

The non-positive definite nature of η allows us to classify tangent vectors $V \in T_p\mathcal{M}$ into three distinct types based on their norm:

- **Time-like:** $\eta(V, V) < 0$. These represent the worldlines of massive particles.
- **Null (Light-like):** $\eta(V, V) = 0$. These represent the paths of massless particles, such as photons.
- **Space-like:** $\eta(V, V) > 0$. These connect events that cannot be causally linked.

In this flat background, the Lorentz Transformation Λ^μ_ν serves as the isometry of the metric, satisfying $\eta_{\alpha\beta} = \eta_{\mu\nu} \Lambda^\mu_\alpha \Lambda^\nu_\beta$. This ensures that the speed of light c remains invariant for all inertial observers.

While Special Relativity assumes a fixed, flat metric where $\partial_\rho \eta_{\mu\nu} = 0$, General Relativity (GR) treats the metric $g_{\mu\nu}$ as a dynamic field. As we will see in the following sections, gravitational waves are essentially small, oscillating perturbations $h_{\mu\nu}$ propagating across this Minkowski background:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1.$$

This “linearized gravity” approach, which we develop in Chapter 2, relies heavily on the index notation and contraction rules established in our preliminaries.

5.2. Intro to General Relativity. General Relativity (GR) is the realization that the “force” of gravity is not a force at all, but rather a consequence of the curvature of the four-dimensional spacetime manifold. Building on the concepts of smooth manifolds and the Levi-Civita connection, GR is founded on two primary pillars: the Equivalence Principle and the Einstein Field Equations.

The physical core of the theory lies in the Equivalence Principle, which posits that at any point in a gravitational field, it is possible to choose a local coordinate system such that the laws of physics take the same form as in an unaccelerated Cartesian system. In mathematical terms, this means that for any point $p \in M$, there exists a coordinate chart such that the metric $g_{\mu\nu}$ at p is exactly the Minkowski metric $\eta_{\mu\nu}$ and its first derivatives (and thus the Christoffel symbols) vanish.

In SR, we assumed a flat background $\eta_{\mu\nu}$ where particles move in straight lines. In GR, the metric $g_{\mu\nu}$ becomes a dynamic variable that encodes the local “refractive” properties of spacetime. The movement of matter is governed by the geometry:

- **Geodesic Motion:** In the absence of external non-gravitational forces, a particle follows a geodesic. As defined in Section 4.3, this is a curve whose tangent vector is parallel transported along itself ($\nabla_{\dot{\gamma}}\dot{\gamma} = 0$).
- **Spacetime Curvature:** The presence of mass-energy produces curvature, quantified by the Riemann tensor $R^\rho_{\sigma\mu\nu}$. This curvature causes initially parallel geodesics to deviate, an effect physically observed as “gravitational attraction”.

The relationship between matter and geometry is summarized by the dictum: “Matter tells spacetime how to curve, and spacetime tells matter how to move.” This is expressed through the Einstein Tensor $G_{\mu\nu}$, which we defined in Section 4.4 as $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$. The divergence-free property of $G_{\mu\nu}$ ($\nabla^\mu G_{\mu\nu} = 0$) is mathematically essential here, as it matches the local conservation of energy and momentum in physics.

By treating the metric as a field, we can eventually analyze how ripples in this geometry—gravitational waves—propagate through the vacuum.

5.3. Intro to the Schwarzschild Metric. The Schwarzschild metric was the first exact solution found for the Einstein field equations, describing the gravitational field outside a spherical, non-rotating, uncharged mass. It provides the mathematical basis for understanding planetary orbits, the bending of light, and the nature of black holes.

In Schwarzschild coordinates (t, r, θ, φ) , the metric tensor $g_{\mu\nu}$ represents a vacuum solution where the energy-momentum tensor is zero. The line element is given by:

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2).$$

We often simplify this using the Schwarzschild radius, defined as $r_s = 2GM/c^2$. Substituting this, we have:

$$ds^2 = - \left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2,$$

where $d\Omega^2 = d\theta^2 + \sin^2(\theta) d\varphi^2$ is the standard metric on a 2-sphere.

Some physical implications of the Schwarzschild metric include:

- **Asymptotic Flatness:** As $r \rightarrow \infty$, the metric components $g_{\mu\nu}$ approach the Minkowski values $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$, meaning gravity vanishes at great distances.
- **The Event Horizon:** A coordinate singularity appears at $r = r_s$. This surface is the event horizon; for a light ray to escape from within this radius, it would require an infinite “acoustic effort” or time-of-flight.
- **Gravitational Redshift:** The g_{tt} component indicates that time (τ) flows more slowly near a massive body compared to a distant observer, a phenomenon called time dilation.

As we discussed in Section 4.3, particles in this geometry follow geodesics. By solving the geodesic equation:

$$\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0,$$

for this specific metric, we can derive the precise orbits of planets and the deflection of light—successes that provided the first experimental proofs of General Relativity.

5.4. Einstein’s Field Equations. The Einstein Field Equations (EFE) are the centerpiece of General Relativity. They provide the precise mathematical dictionary that translates between the geometry of a manifold and the physics of the matter contained within it.

To represent the “matter” side of the equation, we introduce the stress-energy tensor $T_{\mu\nu}$. This is a symmetric $(0, 2)$ -tensor that describes the density and flux of energy and momentum in spacetime. For a perfect fluid, it is written as:

$$T_{\mu\nu} = (\rho + p/c^2)u_\mu u_\nu + pg_{\mu\nu},$$

where ρ is the energy density and p is the isotropic pressure.

The field equations relate the Einstein tensor $G_{\mu\nu}$, which we defined in Section 4.4 as the divergence-free part of the curvature, to the stress-energy tensor. In units where G is the gravitational constant and c is the speed of light, the equations are:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

Here, Λ represents the cosmological constant, which accounts for the energy density of the vacuum.

The mathematical significances of the EFE include:

- **Non-linearity:** Because the Einstein tensor is constructed from the metric and its derivatives, the EFE is a system of ten highly non-linear partial differential equations.
- **Energy Conservation:** As noted in Section 4.4, the contracted Bianchi identity ensures $\nabla^\mu G_{\mu\nu} = 0$. This mathematical requirement forces the physical conservation of energy and momentum: $\nabla^\mu T_{\mu\nu} = 0$.
- **The Source of Waves:** In a vacuum where $T_{\mu\nu} = 0$, the equations reduce to the condition that the Ricci tensor vanishes ($R_{\mu\nu} = 0$). However, as we will explore in Chapter 2, the Weyl tensor $C_{\rho\sigma\mu\nu}$ can still be non-zero, allowing gravitational ripples to propagate through “empty” space.

CHAPTER 2

Mathematical Framework of Gravitational Waves

Now that we’ve developed the necessary mathematical tools and concepts, we can delve into the mathematical framework of gravitational waves.

1. The Linearized Einstein Field Equations

1.1. The Weak-Field Limit and Metric Perturbation. To understand how gravitational waves propagate, we must determine how the metric perturbation $h_{\mu\nu}$ affects the curvature of the manifold. In Chapter 1, we defined the Riemann curvature tensor in terms of Christoffel symbols. Here, we apply the perturbation ansatz $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and discard all terms of $\mathcal{O}(h^2)$.

The Christoffel symbols $\Gamma_{\mu\nu}^\rho$ are defined by derivatives of the metric. In the weak-field limit, the background metric $\eta_{\mu\nu}$ is constant ($\partial\eta = 0$), so the connection is determined solely by the derivatives of the perturbation:

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2}\eta^{\rho\lambda}(\partial_\mu h_{\nu\lambda} + \partial_\nu h_{\mu\lambda} - \partial_\lambda h_{\mu\nu}) + \mathcal{O}(h^2).$$

Notice that the connection itself is first-order in h . Consequently, in any product of connections (like those appearing in the full Riemann tensor formula), the result would be second-order and is therefore neglected.

The Riemann tensor $R_{\sigma\mu\nu}^\rho$ reduces to the difference of the partial derivatives of the connection:

$$R_{\sigma\mu\nu}^\rho = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho.$$

Substituting our linearized connection into this expression yields the linearized Riemann tensor in terms of the metric perturbation:

$$R_{\rho\sigma\mu\nu} = \frac{1}{2}(\partial_\sigma \partial_\mu h_{\rho\nu} + \partial_\rho \partial_\nu h_{\sigma\mu} - \partial_\sigma \partial_\nu h_{\rho\mu} - \partial_\rho \partial_\mu h_{\sigma\nu}).$$

This expression is manifestly invariant under the linearized gauge transformations $h_{\mu\nu} \rightarrow h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu$, confirming that the curvature is a physical observable independent of our choice of coordinates.

By contracting the Riemann tensor ($R_{\sigma\nu} = \eta^{\rho\mu} R_{\rho\sigma\mu\nu}$), we obtain the linearized Ricci tensor:

$$R_{\mu\nu} = \frac{1}{2}(\partial_\sigma \partial_\mu h_\nu^\sigma + \partial_\sigma \partial_\nu h_\mu^\sigma - \partial_\mu \partial_\nu h - \square h_{\mu\nu}),$$

where h is the trace h_μ^μ and $\square = \partial^\sigma \partial_\sigma$ is the d’Alembertian operator. The Ricci scalar is then $R = \eta^{\mu\nu} R_{\mu\nu} = \partial_\mu \partial_\nu h^{\mu\nu} - \square h$.

Combining these, the linearized Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}R$ becomes:

$$G_{\mu\nu} = \frac{1}{2}(\partial_\sigma \partial_\mu h_\nu^\sigma + \partial_\sigma \partial_\nu h_\mu^\sigma - \partial_\mu \partial_\nu h - \square h_{\mu\nu} - \eta_{\mu\nu} \partial_\rho \partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu} \square h).$$

This complex expression is the “geometry” side of the EFE. In the following sections, we will show how the trace-reversal and Lorenz gauge simplify this significant mathematical overhead into a elegant wave equation.

1.2. Linearized Curvature and the Riemann Tensor. To determine the dynamics of the gravitational field in the weak-field limit, we must analyze how the perturbation $h_{\mu\nu}$ generates curvature. In full General Relativity, curvature is a highly non-linear function of the metric; however, the assumption $|h_{\mu\nu}| \ll 1$ allows us to discard higher-order terms, leading to a linear relationship between the perturbation and the Riemann tensor.

We begin with the Levi-Civita connection $\Gamma_{\mu\nu}^\rho$ defined in Section 4.2. In a flat Minkowski background, the connection vanishes. Thus, for $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the Christoffel symbols are determined solely by the first derivatives of $h_{\mu\nu}$:

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2}\eta^{\rho\lambda}(\partial_\mu h_{\nu\lambda} + \partial_\nu h_{\mu\lambda} - \partial_\lambda h_{\mu\nu}) + \mathcal{O}(h^2).$$

Because Γ is already of order $\mathcal{O}(h)$, any product of two connection coefficients—such as those found in the full definition of the Riemann tensor—is of order $\mathcal{O}(h^2)$ and is neglected in the linearization process.

Under these approximations, the Riemann tensor components $R_{\sigma\mu\nu}^\rho$ are reduced to the difference of the partial derivatives of the connection coefficients:

$$R_{\sigma\mu\nu}^\rho \approx \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho.$$

Substituting the expression for the linearized connection and lowering the first index using $\eta_{\rho\lambda}$, we arrive at the linearized Riemann tensor:

$$R_{\rho\sigma\mu\nu} = \frac{1}{2}(\partial_\sigma \partial_\mu h_{\rho\nu} + \partial_\rho \partial_\nu h_{\sigma\mu} - \partial_\sigma \partial_\nu h_{\rho\mu} - \partial_\rho \partial_\mu h_{\sigma\nu}).$$

This expression is remarkably symmetric and, as we will explore in the next section, is invariant under infinitesimal coordinate transformations (gauge transformations).

By contracting the Riemann tensor using the background metric $\eta^{\rho\mu}$, we obtain the linearized Ricci tensor:

$$R_{\sigma\nu} = \eta^{\rho\mu} R_{\rho\sigma\mu\nu} = \frac{1}{2}(\partial_\sigma \partial_\lambda h_\nu^\lambda + \partial_\nu \partial_\lambda h_\sigma^\lambda - \partial_\sigma \partial_\nu h - \square h_{\sigma\nu}),$$

where $h = \eta^{\mu\nu} h_{\mu\nu}$ is the trace and $\square = \partial^\lambda \partial_\lambda$ is the d'Alembertian operator. Finally, the linearized Einstein tensor $G_{\mu\nu}$ is constructed as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} R.$$

In the context of the Einstein vacuum equations ($G_{\mu\nu} = 0$), this expression governs the propagation of $h_{\mu\nu}$.

While standard General Relativity utilizes only the symmetric part of the metric, modern frameworks like Gravitational Quantum Field Theory (GQFT) consider both symmetric and antisymmetric components. In such cases, the Riemann tensor may include additional terms sourced by antisymmetric energy-momentum tensors, potentially leading to the five propagating modes (scalar, vector, and tensor) discussed later in this chapter.

1.3. Gauge Transformations and Invariance. In the full theory of General Relativity, the choice of a coordinate system is arbitrary; the physical laws are invariant under any smooth change of coordinates. When we linearize the theory, this principle of general covariance manifests as a gauge symmetry, very similar to the gauge invariance found in Maxwell's equations for electromagnetism.

Consider a small coordinate transformation of the form:

$$x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu(x),$$

where ξ^μ is a vector field of the same order as the perturbation $h_{\mu\nu}$. Under this transformation, the metric $g_{\mu\nu}$ transforms in a way that induces a change in the perturbation:

$$h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu.$$

The terms $\partial_\mu \xi_\nu + \partial_\nu \xi_\mu$ represent “gauge degrees of freedom”—changes in the components of the metric that arise purely from the choice of coordinates rather than from any physical change in the curvature of spacetime.

A critical result for our framework is that the linearized Riemann tensor $R_{\rho\sigma\mu\nu}$ we derived in Section 2.1.2 is gauge invariant. If we substitute $h'_{\mu\nu}$ into our formula for the Riemann tensor, the terms involving ξ_μ vanish due to the commutativity of partial derivatives:

$$\partial_\alpha \partial_\beta (\partial_\mu \xi_\nu + \partial_\nu \xi_\mu) - \text{permutations} = 0.$$

This confirms that the curvature remains the only truly physical observable. It allows us to distinguish between “physical” waves that actually curve spacetime and “coordinate” waves that can be vanished by a simple change of perspective.

To solve the linearized Einstein Field Equations, we must “fix” the gauge—essentially choosing a coordinate system that makes the math manageable. The most common choice is the Lorenz gauge (or harmonic gauge), which requires:

$$\partial^\mu h_{\mu\nu} = \frac{1}{2} \partial_\nu h.$$

As we will see in 1.5, this choice is mathematically equivalent to the Lorenz gauge in electromagnetism ($\partial_\mu A^\mu = 0$) and allows the complex Einstein tensor to be written as a simple wave operator.

In more advanced frameworks like Gravitational Quantum Field Theory (GQFT), the gauge symmetry must also account for the antisymmetric part of the field. In these cases, the gauge invariance is extended to ensure that the scalar and vector propagating modes are also handled consistently within the unified gravitational field.

1.4. The Trace-Reversed Perturbation. While the metric perturbation $h_{\mu\nu}$ and the linearized Riemann tensor allow us to describe curvature, the resulting Einstein tensor $G_{\mu\nu}$ is still mathematically cumbersome. To streamline the field equations, we define a new tensor field known as the trace-reversed perturbation, denoted as $\bar{h}_{\mu\nu}$.

The trace-reversed perturbation is defined by subtracting half the trace of the original perturbation from itself:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h,$$

where $h = \eta^{\mu\nu} h_{\mu\nu}$ is the trace of the perturbation. The name “trace-reversed” stems from the fact that the trace of this new tensor is the negative of the original trace:

$$\bar{h} = \eta^{\mu\nu} \bar{h}_{\mu\nu} = h - \frac{1}{2} (4) h = -h.$$

The primary utility of $\bar{h}_{\mu\nu}$ is its effect on the linearized Einstein tensor. In terms of the original perturbation $h_{\mu\nu}$, the Einstein tensor $G_{\mu\nu}$ contains several divergent terms. By substituting $\bar{h}_{\mu\nu}$ into the expression for $G_{\mu\nu}$ derived in Section 2.1.2, we obtain a much more symmetrical form:

$$G_{\mu\nu} = -\frac{1}{2} [\Box \bar{h}_{\mu\nu} + \eta_{\mu\nu} \partial^\rho \partial^\sigma \bar{h}_{\rho\sigma} - \partial^\rho \partial_\nu \bar{h}_{\mu\rho} - \partial^\rho \partial_\mu \bar{h}_{\nu\rho}],$$

where $\square = \partial^\rho \partial_\rho$ is the d'Alembertian operator.

This “trace-reversal” trick is a standard technique in the Cauchy problem for the Einstein equations. It essentially prepares the geometry for the application of the Lorenz gauge condition. As we will see in Section 2.1.5, when we require that $\bar{h}_{\mu\nu}$ be divergence-free ($\partial^\mu \bar{h}_{\mu\nu} = 0$), all the complex partial derivative terms in the brackets vanish, leaving behind only the wave operator.

In the context of Gravitational Quantum Field Theory (GQFT), the trace-reversed perturbation remains a vital component. However, in these extended theories, the symmetric part of the energy-momentum tensor specifically sources the trace and traceless parts of the quadrupole moment, which are directly related to the scalar and tensor components of $\bar{h}_{\mu\nu}$.

1.5. Lorenz Gauge and the Vacuum Wave Equation. The primary goal of the linearized framework is to simplify the complex, non-linear Einstein Field Equations into a form that describes wave propagation. By combining the trace-reversed perturbation $\bar{h}_{\mu\nu}$ with a specific choice of coordinate system, we can reduce the Einstein tensor to the standard d'Alembertian operator.

As established in Section 1.3, we possess the gauge freedom to perform infinitesimal coordinate transformations without altering the physical curvature. To simplify the field equations, we impose the Lorenz gauge condition (also known as the harmonic or Hilbert gauge):

$$\partial^\mu \bar{h}_{\mu\nu} = 0.$$

This condition is mathematically analogous to the Lorenz gauge in electromagnetism ($\partial_\mu A^\mu = 0$). Physically, it restricts our coordinate system to one where the “coordinates” themselves move in a way that respects the wave-like nature of the perturbation.

In Section 1.4, we saw that the linearized Einstein tensor in terms of the trace-reversed perturbation is:

$$G_{\mu\nu} = -\frac{1}{2} \left[\square \bar{h}_{\mu\nu} + \eta_{\mu\nu} \partial^\rho \partial^\sigma \bar{h}_{\rho\sigma} - \partial^\rho \partial_\nu \bar{h}_{\mu\rho} - \partial^\rho \partial_\mu \bar{h}_{\nu\rho} \right].$$

When the Lorenz gauge condition $\partial^\mu \bar{h}_{\mu\nu} = 0$ is applied, all terms involving the divergence of $\bar{h}$ vanish identically. The vacuum Einstein equations ($G_{\mu\nu} = 0$) then simplify to the vacuum wave equation:

$$\square \bar{h}_{\mu\nu} = 0,$$

where $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$ is the d'Alembertian operator.

The existence of this wave equation proves that perturbations in the geometry of spacetime propagate through the vacuum as waves at the speed of light c . This provides the mathematical foundation for the concept of gravitational radiation: ripples in the fabric of spacetime that carry energy and information away from their sources.

In the framework of Gravitational Quantum Field Theory (GQFT), the wave equation is generalized to include both symmetric and antisymmetric fields. While standard General Relativity results in only tensor waves, the unified field equations in GQFT reveal that the vacuum can also support the propagation of scalar and vector modes, provided the appropriate gauge conditions are met for both the symmetric and antisymmetric components of the gravitational field.

1.6. Sourcing the Waves: The Energy-Momentum Tensor. In the previous subsections, we established that perturbations of spacetime can propagate as waves through a vacuum. However, to understand the origin of these waves, we must look to the “source” side of the Einstein Field Equations: the energy-momentum tensor $T_{\mu\nu}$. This tensor describes the distribution and flow of mass, energy, and momentum throughout the manifold.

When matter is present, the linearized Einstein equations do not vanish. Instead, they relate the d'Alembertian of the trace-reversed perturbation directly to the energy-momentum tensor:

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}.$$

This is an inhomogeneous wave equation. Mathematically, it implies that any time-varying distribution of energy-momentum will act as a source for gravitational waves, much like accelerating charges generate electromagnetic radiation.

A crucial constraint in this framework is the local conservation of energy and momentum, expressed as $\partial^\mu T_{\mu\nu} = 0$. This conservation law has profound implications for the nature of the waves produced:

- **Monopole Radiation:** Because total mass-energy is conserved, there is no gravitational monopole radiation (analogous to the pulsating of a spherical mass).
- **Dipole Radiation:** Because linear and angular momentum are conserved, there is no gravitational dipole radiation.
- **Quadrupole Radiation:** The leading order for gravitational radiation is the quadrupole. Mathematically, waves are generated by the second time derivative of the “quadrupole moment” of the mass distribution, representing asymmetric accelerations like those in a binary star system.

In standard General Relativity, only the symmetric part of the energy-momentum tensor is considered, leading to the two standard tensor polarizations. However, in the Gravitational Quantum Field Theory (GQFT) framework:

- **Symmetric Source ($T_{(\mu\nu)}$):** Generates the two tensor modes and a scalar mode through the trace and traceless parts of the quadrupole moment.
- **Antisymmetric Source ($T_{[\mu\nu]}$):** Can act as a source for the two vector modes of gravitational waves, often associated with the spin of particles like fermions within the source.

To find the perturbation produced by a specific source, we use the method of Green's functions to find the retarded potential:

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4} \int \frac{T_{\mu\nu}(t - |\mathbf{x} - \mathbf{y}|/c, \mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d^3y.$$

This formula shows that the wave observed at a distance is a result of the source's behavior at an earlier “retarded” time, accounting for the finite speed of gravity.

2. Propagating Modes and Polarizations

2.1. The Transverse-Traceless (TT) Gauge. Even after imposing the Lorenz gauge condition $\partial^\mu \bar{h}_{\mu\nu} = 0$, the coordinate system is not uniquely fixed. There exists a residual gauge freedom—infinitesimal transformations $x^\mu \rightarrow x^\mu + \xi^\mu$ —where the gauge vector ξ^μ itself satisfies the wave equation $\square \xi^\mu = 0$. By strategically utilizing this remaining freedom, we can move into the Transverse-Traceless (TT)

gauge, which filters out all non-physical degrees of freedom and leaves only the radiative components of the gravitational wave.

In the TT gauge, the metric perturbation $h_{\mu\nu}$ is subjected to several strict algebraic and differential conditions:

- (1) **Purely Spatial:** We set $h_{0\mu} = 0$, meaning the wave has no components involving the time index.
- (2) **Traceless:** The trace of the perturbation vanishes, $h = \eta^{\mu\nu} h_{\mu\nu} = 0$. A significant consequence of this is that the distinction between the original perturbation and the trace-reversed version disappears: $h_{\mu\nu}^{TT} = \bar{h}_{\mu\nu}^{TT}$.
- (3) **Transverse:** The perturbation is divergence-free in its spatial indices, $\partial^j h_{ij} = 0$. For a wave propagating in the z -direction, this implies that only the x and y components of the tensor can be non-zero.

The TT gauge is the most “physical” frame for several reasons:

- (1) **Coordinate Stability:** In this gauge, the coordinates of a test mass at rest do not change as the wave passes. The “motion” of the particles we observe is actually the stretching and squeezing of the underlying spacetime geometry between those fixed coordinate points.
- (2) **Non-Zero Components:** For a wave traveling along the x^3 (or z) axis, the TT constraints reduce the ten independent components of the symmetric 4×4 matrix to just two:

$$h_{ij}^{TT} = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

These two remaining degrees of freedom, h_+ and $h_\times$, represent the two physical polarizations of the gravitational wave.

While the TT gauge effectively isolates the two tensor modes in General Relativity, the GQFT framework introduces a more complex field containing both symmetric and antisymmetric components. In this theory, the gauge-fixing process must account for the fact that the unified gravitational field possesses five propagating modes. While h_+ and $h_\times$ remain the primary tensor modes, the additional scalar and vector modes require their own specific gauge treatments to ensure they are physically distinguishable from the tensor radiation.

2.2. Classical Tensor Modes: Plus (+) and Cross (×). In the Transverse-Traceless gauge, the gravitational perturbation h_{ij}^{TT} for a wave propagating in the z -direction is reduced to only two independent components. These components correspond to the two polarization states of the gravitational wave, conventionally denoted as plus (+) and cross (×).

For a monochromatic plane wave traveling along the x^3 axis, the spatial components of the perturbation tensor are expressed as:

$$h_{ij}^{TT}(t, z) = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix} \cos(\omega(t - z/c)).$$

Here, h_+ and $h_\times$ are the amplitudes of the two modes. The structure of this matrix reveals the physical requirements of the TT gauge: it is symmetric, its trace is zero ($h_{11} + h_{22} = h_+ - h_+ = 0$), and it is purely transverse to the z -direction of propagation.

The plus polarization affects the proper distance between test particles in a “quadrupolar” fashion. If we consider a ring of test particles in the $x - y$ plane:

- During the first half-cycle, the ring is stretched along the x -axis and simultaneously squeezed along the y -axis.
- During the second half-cycle, the effect reverses: it is squeezed along the x -axis and stretched along the y -axis.

The cross polarization has an identical effect but is rotated by 45° relative to the plus mode.

- It stretches and squeezes the ring along the axes defined by $y = x$ and $y = -x$.
- Mathematically, this corresponds to the off-diagonal components $h_{12} = h_{21}$ in the perturbation tensor.

Because the linearized Einstein equations are linear, any arbitrary gravitational wave can be described as a linear superposition of these two basis states:

$$h_{ij}^{TT} = h_+ e_{ij}^+ + h_\times e_{ij}^\times,$$

where e_{ij}^+ and $e_{ij}^\times$ are the unit polarization tensors. This decomposition is essential for detection, as the orientation of an interferometer relative to the source determines the ratio of h_+ to $h_\times$ that is measured.

While these two tensor modes are the only propagating degrees of freedom in General Relativity, the GQFT framework expands this view. In GQFT, these tensor modes are specifically generated by the traceless part of the symmetric energy-momentum quadrupole moment. However, they are accompanied by additional scalar and vector modes that arise from different components of the unified gravitational field, which we will examine in Section 2.4.

2.3. Helicity and Spin-2 Properties. The classification of gravitational waves as spin-2 fields is rooted in how the metric perturbation h_{ij}^{TT} transforms under rotations around the direction of propagation. This property distinguishes gravity from electromagnetism (a spin-1 theory) and scalar fields (spin-0).

Consider a monochromatic gravitational wave propagating along the z -axis. If we rotate our coordinate system by an angle ψ around this axis, the polarization states h_+ and $h_\times$ do not remain invariant; instead, they transform into a linear combination of one another:

- $h'_+ = h_+ \cos(2\psi) + h_\times \sin(2\psi)$
- $h'_\times = -h_+ \sin(2\psi) + h_\times \cos(2\psi)$

Mathematically, these transformation rules indicate that the field has helicity states of ± 2 . A rotation of $\psi = 180^\circ$ (or π radians) returns the polarization tensor to its original state ($2\psi = 360^\circ$). This is the hallmark of a spin-2 field; in contrast, a spin-1 vector field (like a photon) requires a full 360° rotation to return to its initial state.

The spin-2 nature is a direct consequence of the tensor’s rank and symmetry. Because the metric is a symmetric rank-2 tensor, its radiative degrees of freedom must exhibit this 180° rotational symmetry. Physically, this corresponds to the “quadrupolar” nature of the wave: the stretching and squeezing pattern repeats twice during a single full rotation of the viewing plane.

In the Gravitational Quantum Field Theory (GQFT) framework, the spin properties of the gravitational field are expanded beyond the standard tensor modes:

- **Tensor Modes:** Maintain the standard spin-2 (helicity ± 2) characteristics derived from the symmetric part of the field.
- **Vector Modes:** Exhibit spin-1 (helicity ± 1) properties, arising from the antisymmetric components of the gravitational field.
- **Scalar Mode:** Exhibits spin-0 (helicity 0) properties, representing a longitudinal breathing mode that is invariant under any rotation around the axis of propagation.

This multi-spin structure in GQFT arises because the theory unifies the metric with additional degrees of freedom, suggesting that a complete quantum description of gravity may involve a spectrum of helicity states rather than just the classical spin-2 graviton.

2.4. Beyond General Relativity: Scalar and Vector Modes. While General Relativity (GR) restricts the propagation of gravity to two transverse-traceless tensor modes, the GQFT framework reveals a richer structure of five physical degrees of freedom. In this theory, the gravitational field $G_{\mu\nu}$ is composed of both symmetric and antisymmetric components, which decouple in the linearized vacuum to produce one scalar mode, two vector modes, and the two standard tensor modes.

The scalar mode, often referred to as a “breathing” or “longitudinal” mode, arises from the trace part of the symmetric gravitational field.

- **Mathematical Form:** For a wave propagating in the z -direction, the scalar perturbation $h_{ij}^{(s)}$ affects the diagonal components of the spatial metric equally: $h_{11} = h_{22} \propto \Phi$, where Φ is a scalar potential.
- **Physical Effect:** Unlike the quadrupolar tensor modes that stretch one axis while squeezing the other, the scalar mode causes a symmetric expansion and contraction of the plane transverse to the wave’s motion.
- **Sourcing:** This mode is uniquely generated by the trace of the symmetric energy-momentum tensor, $T = \eta^{\mu\nu} T_{\mu\nu}$, representing isotropic changes in the source’s mass-energy distribution.

The GQFT framework introduces two vector modes—denoted as W_1 and W_2 —which arise from the antisymmetric component of the gravitational field $G_{[\mu\nu]}$.

- **Helicity ± 1 :** These modes transform under rotations like electromagnetic waves (spin-1), requiring a full 360° rotation to return to their original state.
- **Physical Effect:** Vector modes induce a “dipolar” shift in test masses. They are transverse to the direction of propagation but, unlike tensor modes, they tend to displace particles in a way that suggests a rotation or “frame-dragging” effect as the wave passes.
- **Sourcing:** These modes are sourced by the antisymmetric part of the energy-momentum tensor $T_{[\mu\nu]}$, which is physically linked to the intrinsic spin and angular momentum of particles within the source.

By accounting for both symmetric and antisymmetric energy-momentum tensors, the field equations in GQFT provide a complete set of solutions that generalize the Einsteinian view:

- **Tensor Modes ($h_+, h_\times$):** Sourced by the traceless symmetric quadrupole moment.

- **Scalar Mode (Φ):** Sourced by the trace of the symmetric energy-momentum tensor.
- **Vector Modes (W_1, W_2):** Sourced by the antisymmetric energy-momentum tensor.

This expanded classification is not merely a mathematical curiosity; it provides a framework for testing the limits of General Relativity. If a future detector network identifies a “breathing” signal or a vector-like displacement, it would provide direct evidence for the unified field structures predicted by GQFT.

2.5. Physical Effects on Matter: The Geodesic Deviation Perspective.

The ultimate proof of a gravitational wave’s existence lies in its interaction with matter. While the Transverse-Traceless (TT) gauge provides a convenient mathematical coordinate system where test masses appear “stationary,” the physical distance between them is constantly changing. To quantify this, we utilize the Geodesic Deviation Equation, which describes how the curvature of spacetime—represented by the Riemann tensor—pushes and pulls neighboring geodesics.

As established in Section 4.3, a particle in free-fall follows a geodesic. If we have two such particles with a separation vector S^μ , their relative acceleration is governed by the Riemann curvature tensor:

$$\frac{D^2 S^\mu}{d\tau^2} = R^\mu_{\nu\rho\sigma} U^\nu U^\rho S^\sigma.$$

In the weak-field limit, for a detector at rest ($U^\nu \approx \delta_0^\nu$), this equation simplifies significantly. The relative acceleration is directly proportional to the second time derivative of the metric perturbation:

$$\frac{d^2 S^i}{dt^2} \approx \frac{1}{2} \ddot{h}_j^i S^j.$$

This shows that the gravitational wave acts as a tidal force, stretching and squeezing the space between the masses.

The nature of the displacement depends entirely on which modes are present in the wave:

- **Tensor Modes ($h_+, h_\times$):** These produce the characteristic “quadrupolar” deformation. A ring of particles will stretch in one transverse direction while squeezing in the other, always maintaining a constant area in the plane of the ring.
- **Scalar “Breathing” Mode:** Predicted by GQFT, this mode causes the ring to expand and contract symmetrically in all transverse directions at once. This “breathing” effect changes the local volume, a phenomenon strictly forbidden in standard vacuum General Relativity.
- **Vector Modes:** These induce a shifting motion that can be thought of as a “dipolar” or frame-dragging effect. They displace test masses in a way that depends on the direction of propagation, often involving longitudinal components that are not present in pure tensor waves.

This geodesic deviation is exactly what laser interferometers like LIGO are designed to measure. By monitoring the phase shift of laser light traveling between mirrors (test masses), we can reconstruct the components of h_{ij} . However, as highlighted in the GQFT framework, a standard L-shaped interferometer is primarily sensitive to tensor modes. Detecting scalar or vector modes would require observing

specific “in-phase” arm movements or utilizing a global network of detectors to perform a full “polarization resolve” of the incoming signal.

3. Sources of Gravitational Radiation

3.1. The Multipole Expansion of the Field. To understand how physical systems generate gravitational radiation, we must solve the inhomogeneous wave equation derived in Section 1.6: $\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}$. The standard mathematical approach is to use the method of Green’s functions, which yields the retarded potential solution:

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4} \int \frac{T_{\mu\nu}(t - |\mathbf{x} - \mathbf{y}|/c, \mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d^3y.$$

For a distant observer at a distance $R = |\mathbf{x}|$, where the source size d satisfies $d \ll R$, we can perform a multipole expansion of this integral.

In electromagnetism, the leading order of radiation is the dipole. However, in gravitation, the conservation laws established by the Einstein Field Equations impose stricter constraints:

- **Monopole (Mass):** The $1/R$ term in the expansion corresponds to the total mass-energy of the system. Since mass-energy is conserved ($\partial_\mu T^{\mu 0} = 0$), the monopole moment does not vary in time, and thus produces no radiation.
- **Dipole (Momentum):** The next term relates to the first mass moment (center of mass) and the current dipole (angular momentum). Conservation of linear and angular momentum ($\partial_\mu T^{\mu i} = 0$) ensures these moments have vanishing second time derivatives, precluding gravitational dipole radiation.
- **Quadrupole:** The first non-vanishing radiative term appears at the $1/c^4$ order, corresponding to the quadrupole moment. This represents the leading order of gravitational wave generation in General Relativity.

In the “far-field” or wave zone, we assume the source is “slow-motion” ($v \ll c$), meaning the wavelength λ is much larger than the source size d . Under these conditions, the spatial components of the perturbation can be approximated purely by the second time derivative of the source’s mass distribution:

$$\bar{h}_{ij}(t, R) \approx \frac{2G}{c^4 R} \frac{d^2 I_{ij}}{dt^2} (t - R/c),$$

where I_{ij} is the second moment of the mass distribution.

In the GQFT framework, the expansion is more complex due to the presence of both symmetric and antisymmetric energy-momentum tensors.

- **Symmetric Part ($T_{(\mu\nu)}$):** Continues to source the standard tensor waves via the traceless quadrupole moment and scalar waves via the trace of the quadrupole moment.
- **Antisymmetric Part ($T_{[\mu\nu]}$):** Introduces “spin-current” multipoles. While mass-dipole radiation remains forbidden, the time-varying intrinsic spin distribution of the source can generate vector-type gravitational waves at the same order as the standard quadrupole radiation.

3.2. The Quadrupole Formula in General Relativity. In General Relativity, the quadrupole formula is the primary tool for calculating the gravitational radiation produced by a system in slow, non-relativistic motion ($v \ll c$). Because the monopole and dipole moments are stationary due to conservation laws, the first

time-varying component of the gravitational field arises from the mass quadrupole moment.

We define the mass quadrupole moment tensor, I_{ij} , as the second moment of the mass distribution:

$$I_{ij}(t) = \int \rho(t, \mathbf{x}) x_i x_j d^3x,$$

where ρ is the mass density of the source. To isolate the parts of the distribution that actually contribute to radiation, we often use the reduced (traceless) quadrupole moment, Q_{ij} :

$$Q_{ij} = I_{ij} - \frac{1}{3} \delta_{ij} I_k^k.$$

This tensor Q_{ij} describes the degree of asymmetry in the mass distribution; a perfectly spherical object has a vanishing reduced quadrupole moment and thus radiates no gravitational waves.

In the far-field limit ($R \gg d$), the spatial components of the trace-reversed perturbation $\bar{h}_{ij}$ are proportional to the second time derivative of the quadrupole moment, evaluated at the retarded time $t_{ret} = t - R/c$:

$$\bar{h}_{ij}(t, R) = \frac{2G}{c^4 R} \ddot{Q}_{ij}(t - R/c).$$

This formula illustrates why gravitational waves are so difficult to detect: the factor G/c^4 is approximately $8.2 \times 10^{-45} \text{s}^2/\text{kg} \cdot \text{m}$, meaning even massive astrophysical accelerations produce only miniscule strains in spacetime.

The quadrupole formula also allows us to calculate the luminosity (P), or the rate at which the system loses energy to gravitational radiation:

$$P = \frac{G}{5c^5} \left\langle \ddot{Q}_{ij} \ddot{Q}^{ij} \right\rangle,$$

where the brackets denote an average over several wavelengths. This energy loss is what causes binary systems, like the Hulse-Taylor pulsar, to slowly spiral inward as their orbital energy is carried away by waves.

In the GQFT framework, this classical result is preserved as the leading order for tensor waves. However, GQFT clarifies that the symmetric energy-momentum tensor $T_{(\mu\nu)}$ is responsible for this coupling. While the traceless part of the quadrupole moment produces the standard h_+ and $h_\times$ tensor modes, the trace of the quadrupole moment can source the scalar (breathing) mode, which is missing from the standard Einsteinian derivation.

3.3. Antisymmetric Sources and Spin-Generated Waves. While standard General Relativity focuses almost exclusively on the symmetric part of the energy-momentum tensor, the Gravitational Quantum Field Theory (GQFT) framework explicitly incorporates the antisymmetric energy-momentum tensor, $T_{[\mu\nu]}$, as a fundamental source of gravitational radiation. This addition expands the physical landscape of gravitational waves to include modes generated by the intrinsic angular momentum, or spin, of the matter within a source.

In GQFT, the unified gravitational field is governed by equations where both the symmetric and antisymmetric parts of the field are coupled to their respective parts of the energy-momentum tensor. Specifically, the antisymmetric part of the field equations can be expressed as:

$$\square f_{\mu\nu} = -16\pi G T_{[\mu\nu]},$$

where $f_{\mu\nu}$ represents the antisymmetric gravitational field. This demonstrates that if a system possesses a time-varying antisymmetric stress-energy distribution, it will radiate gravitational energy that is mathematically distinct from the standard quadrupole radiation.

The most significant physical consequence of antisymmetric sources is the generation of vector gravitational waves.

- **Physical Source:** These waves are sourced by the “spin-current” of the system. For example, a system of fermions with time-varying spin orientations or a rotating body with a fluctuating distribution of intrinsic angular momentum would act as a source.
- **Dipolar Nature:** Unlike the mass-quadrupole radiation which is tensor-based, spin-generated radiation can exhibit a dipolar structure. While the mass-dipole is forbidden by the conservation of linear momentum, the “spin-dipole” is allowed because it relates to the internal distribution of angular momentum rather than the net displacement of the center of mass.

It is also important to note that within this unified framework, the symmetric energy-momentum tensor $T_{(\mu\nu)}$ is responsible for both tensor and scalar modes. While the traceless part sources the standard h_+ and $h_\times$ waves, the trace of the symmetric energy-momentum tensor sources the scalar (breathing) mode:

$$\square\Phi \propto \eta^{\mu\nu}T_{(\mu\nu)}.$$

This means that a source undergoing isotropic pulsations (changes in volume or pressure) will radiate scalar gravitational waves, a phenomenon that does not occur in vacuum General Relativity.

The total gravitational radiation from a complex source in GQFT is the sum of contributions from three distinct mathematical “channels”:

- **Traceless Symmetric Quadrupole:** Sources the two tensor modes ($h_+, h_\times$).
- **Trace of Symmetric Tensor:** Sources the scalar breathing mode (Φ).
- **Antisymmetric Spin-Current:** Sources the two vector modes (W_1, W_2).

3.4. Energy and Momentum Carried by Waves. Although gravitational waves are perturbations of the spacetime metric, they are not merely geometric abstractions; they carry physical energy, momentum, and angular momentum away from their sources. This energy transport is a non-linear effect of General Relativity, as the waves themselves contribute to the curvature of the spacetime through which they propagate.

Because the Einstein Field Equations are non-linear, the energy of the gravitational field cannot be localized at a single point. Instead, we define the energy-momentum carried by the waves using the Isaacson stress-energy tensor, $t_{\mu\nu}^{GW}$. This tensor is calculated by averaging the products of the derivatives of the perturbation $h_{\mu\nu}$ over several wavelengths:

$$t_{\mu\nu}^{GW} = \frac{c^4}{32\pi G} \left\langle \partial_\mu h_{ij}^{TT} \partial_\nu h_{TT}^{ij} \right\rangle.$$

This “effective” stress-energy tensor acts as a source in the second-order Einstein equations, representing the gravitational “back-reaction” on the background metric.

The amount of energy passing through a unit area per unit time (the flux) is determined by the t_{0i}^{GW} components. By integrating this flux over a large sphere

surrounding the source, we derive the total luminosity (P), which for a standard quadrupole source is:

$$P = \frac{G}{5c^5} \left\langle \ddot{Q}_{ij} \ddot{Q}^{ij} \right\rangle.$$

This formula shows that the power radiated is proportional to c^{-5} , explaining why only the most violent astrophysical events—such as merging black holes—produce detectable levels of radiation.

In addition to energy, gravitational waves carry:

- **Linear Momentum:** If a source radiates asymmetrically, it can experience a “gravitational kick,” potentially ejecting a merged black hole from its host galaxy.
- **Angular Momentum:** Waves from rotating systems carry away orbital angular momentum, which is the primary driver for the orbital decay observed in binary pulsars.

The Gravitational Quantum Field Theory (GQFT) framework extends this by showing that the energy is distributed across all five propagating modes.

- **Tensor Channels:** Carry the bulk of the energy in standard asymmetric mergers.
- **Scalar and Vector Channels:** In systems with high intrinsic spin or isotropic volume changes, a significant fraction of the total radiated energy can be carried away by the scalar “breathing” mode or the vector “spin” modes.
- **Unified Conservation:** GQFT ensures that the total energy lost by the combined symmetric and antisymmetric energy-momentum tensors matches the energy flux detected in the unified gravitational field $G_{\mu\nu}$.

3.5. Classification of Astrophysical Sources. To conclude this section, we classify the physical systems that generate detectable gravitational radiation. These sources are generally categorized by the duration and characteristics of their signal, which is determined by the mathematical properties of their mass-energy and spin-current distributions.

The most prominent sources are binary systems composed of compact objects, such as black holes (BH) or neutron stars (NS). The signal is divided into three mathematical phases:

- **Inspiral:** As the objects orbit each other, they lose energy via the quadrupole formula, causing the orbital frequency and amplitude to increase—a phenomenon known as a “chirp”.
- **Merger:** The non-linear regime where the two objects collide. This phase requires numerical relativity to solve the full Einstein Field Equations.
- **Ringdown:** The newly formed single object vibrates as it settles into a stable state (e.g., a Kerr black hole), emitting damped oscillations.

Continuous Sources are long-lived, nearly monochromatic signals produced by objects with a persistent, time-varying quadrupole moment.

- **Asymmetric Neutron Stars:** A rapidly spinning neutron star with a small “mountain” or deformation on its surface will radiate continuously.
- **Spin-Current Generation:** According to GQFT, if such a star has a fluctuating intrinsic spin distribution, it could also emit continuous vector-type gravitational waves.

Bursts are short-duration, “unmodeled” signals from sudden, violent events.

- **Supernovae:** The core collapse of a massive star produces a burst of radiation. The signal’s strength depends heavily on the degree of asymmetry in the explosion.
- **Collapsars:** As identified in recent 2024 studies, the deaths of rapidly spinning stars can emit coherent bursts that provide a window into the star’s internal collapse.

The Stochastic Background is a “hum” of gravitational radiation consisting of the superposition of countless unresolved sources.

- **Cosmological Background:** Waves originating from the Big Bang or phase transitions in the early universe.
- **Astrophysical Background:** The combined noise of all the binary mergers throughout the history of the universe that are too distant to be identified individually.

In the context of GQFT, these sources are not limited to tensor radiation. For example, a “breathing” binary or a pulsar with significant spin-current fluctuations would contribute to the scalar and vector backgrounds, respectively. Distinguishing these modes in the signals from the sources listed above remains one of the primary goals of next-generation gravitational wave astronomy.

4. Detection and Geodesic Deviation

4.1. The Geodesic Deviation Equation. In General Relativity, a single test mass in free-fall follows a geodesic, the curved-space equivalent of a straight line. To detect a gravitational wave, however, we cannot look at one particle in isolation; we must observe the relative motion between two or more test masses. The mathematical framework for this relative motion is the Equation of Geodesic Deviation, also known as the Jacobi equation.

Consider two nearby particles moving along adjacent geodesics. Let $U^\mu = dx^\mu/d\tau$ be the four-velocity of the first particle and S^μ be the separation vector pointing from the first particle to the second. As the particles move through a region of curved spacetime, the acceleration of their separation is governed by the Riemann curvature tensor $R^\mu_{\nu\rho\sigma}$:

$$(2.1) \quad \frac{D^2 S^\mu}{d\tau^2} = R^\mu_{\nu\rho\sigma} U^\nu U^\rho S^\sigma$$

This equation demonstrates that “tidal forces”—the forces that stretch and squeeze objects—are not a separate physical entity but are the direct manifestation of space-time curvature.

To apply this to a detector like LIGO, we simplify Equation 2.1 using the assumptions of the linearized theory and the proper detector frame (a coordinate system tied to the center of the detector):

- **Non-Relativistic Limit:** For test masses that are nearly stationary before the wave arrives, $U^0 \approx 1$ and $U^i \approx 0$.
- **Linearized Riemann** In the Transverse-Traceless (TT) gauge, the components of the Riemann tensor are directly related to the second time derivative of the metric perturbation: $R^i_{0j0} = -\frac{1}{2}\ddot{h}^{TT}_{ij}$.

Substituting these into the Jacobi equation yields the fundamental detection formula:

$$\ddot{S}^i = \frac{1}{2} \ddot{h}_{ij}^{TT} S^j.$$

This shows that the acceleration of the test masses is proportional to the amplitude of the gravitational wave and the original distance between the masses.

In the GQFT framework, the geodesic deviation becomes a more powerful diagnostic tool because different modes affect the separation vector S^μ in distinct ways:

- **Tensor Deviation:** Causes the standard differential change in length between perpendicular arms (e.g., S^x stretches while S^y shrinks).
- **Scalar Deviation:** Induces a symmetric “breathing” effect where both S^x and S^y increase or decrease simultaneously.
- **Vector Deviation:** Can induce longitudinal displacements (S^z) or “frame-dragging” rotations, depending on the orientation of the spin-current source.

By analyzing the specific pattern of geodesic deviation across a network of detectors, researchers can determine not just the strength of the wave, but which of the five physical modes are present.

4.2. Interaction with Interferometers. While the geodesic deviation equation describes how test masses should move, the displacements caused by gravitational waves are incredibly small—typically on the order of 10^{-18} meters, or one-thousandth the diameter of a proton. To measure such infinitesimal changes, experimental physics utilizes Michelson interferometry.

A standard gravitational wave interferometer, such as LIGO or Virgo, consists of two perpendicular arms of length L . A laser beam is split at a corner, travels down both arms, reflects off suspended mirrors (the test masses), and returns to the beam splitter to recombine.

- **Destructive Interference:** The detector is typically tuned so that, in the absence of a wave, the light from the two arms cancels out at the photodetector, resulting in “darkness”.
- **Differential Phase Shift:** When a gravitational wave passes, the metric perturbation $h_{\mu\nu}$ alters the proper distance of the arms differently. For a + polarized wave, one arm stretches ($L + \Delta L$) while the other squeezes ($L - \Delta L$).

Now, let’s calculate the Phase Shift. The time it takes for light to travel the round-trip distance in an arm is affected by the metric. For an arm along the x -axis, the interval $ds^2 = 0$ for light implies:

$$c^2 dt^2 = (1 + h_{xx}) dx^2.$$

Integrating this over the length of the arm shows that the change in the round-trip time Δt is proportional to the amplitude of the wave h . This time delay manifests as a phase shift $\Delta\varphi$ in the laser light:

$$\Delta\varphi = \frac{4\pi L}{\lambda} h(t).$$

where λ is the wavelength of the laser. This shift causes the light to no longer interfere destructively, allowing a signal to reach the photodetector.

The “antenna response” of an interferometer depends on its orientation relative to the wave.

- **Tensor Sensitivity:** An L-shaped detector is most sensitive to tensor waves arriving from directly “above” the plane of the arms.
- **Scalar Sensitivity:** As noted in the GQFT framework, a scalar “breathing” mode would cause both arms to stretch and squeeze in unison. In a perfectly symmetric Michelson interferometer, this would result in no net phase shift, making the detector “blind” to certain scalar signals unless the arms are of unequal length or multiple detectors are cross-correlated.

To increase the effective arm length L without building thousand-mile-long tunnels, modern detectors use Fabry-Pérot cavities. By placing mirrors near the beam splitter, the light is reflected back and forth roughly 300 times before being recombined. This multiplies the phase shift, allowing the detector to reach the extreme sensitivity required to observe the distant “chirps” of merging black holes.

4.3. Measuring Non-Standard Modes in GQFT. In General Relativity, the existence of only two tensor polarizations $(+, \times)$ is a rigid prediction. However, as derived in Section 2.2.4, the GQFT framework predicts a total of five propagating modes: two tensor, two vector, and one scalar. Detecting these “non-standard” modes would provide the first direct evidence of physics beyond the Einsteinian metric.

To distinguish between these modes, we must analyze the antenna response function F of an interferometer. The response of a detector to a specific mode is determined by the contraction of the detector’s geometry tensor D^{ij} with the wave’s polarization tensor e_{ij} :

$$h(t) = F_+ h_+ + F_\times h_\times + F_x W_1 + F_y W_2 + F_s \Phi.$$

Because each mode (tensor, vector, or scalar) has a different geometric footprint, each will “illuminate” an L-shaped interferometer differently depending on the wave’s direction of arrival.

A single L-shaped interferometer (like LIGO Hanford) cannot distinguish a scalar breathing mode from a tensor mode on its own; it only measures a single time-varying strain $h(t)$. To “disentangle” the five modes, we require a global network of detectors—including LIGO (USA), Virgo (Italy), KAGRA (Japan), and LIGO-India.

- **Triangulation:** By comparing the arrival times at different sites, we can determine the exact direction of the source.
- **Polarization Resolving:** Because each detector is oriented differently on the Earth’s curved surface, a single wave will produce a unique combination of strains across the network. By solving the resulting system of linear equations, researchers can mathematically isolate the amplitudes of the h_+ , $h_\times$, W_1 , W_2 , and Φ components.

One of the most sought-after signatures in GQFT is the scalar breathing mode (Φ).

- **Detection Signature:** In a perfectly symmetric interferometer, a scalar wave causes both arms to stretch and squeeze in-phase, resulting in zero net phase shift.
- **Non-Null Detection:** To detect it, physicists look for “null stream” violations—signals that appear in a way that tensor waves cannot explain—or utilize detectors with arms of unequal lengths or different orientations.

Recent analyses of binary black hole mergers (such as GW170814) have allowed for the first tests of these non-standard polarizations. While current data remains

strongly consistent with pure tensor modes, the precision of the GQFT framework provides a rigorous “null hypothesis” that will be tested with much higher sensitivity by next-generation detectors like the Einstein Telescope and LISA.

4.4. Geodesic Deviation in the “Memory Effect”. In the standard treatment of gravitational waves, we assume that after a wave passes, the spacetime returns exactly to its original Minkowski state. However, the non-linear nature of the Einstein Field Equations predicts a memory effect—a permanent, non-zero displacement in the separation vector S^i between test masses long after the wave has vanished.

The memory effect is mathematically split into two distinct components:

- **Linear Memory:** Arises from the change in the velocities of the source’s components (e.g., a burst of particles flying off to infinity).
- **Nonlinear (Christodoulou) Memory:** A purely general-relativistic effect where the gravitational waves themselves act as a source of gravity. As the waves carry energy-momentum to “null infinity,” they create a permanent “warp” in the local spacetime metric.

Recall the geodesic deviation equation $\ddot{S}^i = \frac{1}{2}\ddot{h}_{ij}^{TT}S^j$. Integrating this equation twice over the duration of a gravitational wave event reveals the change in separation ΔS^i :

$$\Delta S^i = \frac{1}{2}\Delta h_{ij}^{TT}S^j,$$

where Δh_{ij}^{TT} is the difference between the metric perturbation at the very beginning and the very end of the signal. In a system with “memory,” the metric does not return to its initial value ($h_{final} \neq h_{initial}$), meaning the test masses are left in a permanently shifted configuration.

The GQFT framework provides a unique perspective on the memory effect by linking it to the different propagating modes:

- **Tensor Memory:** The standard permanent “stretch and squeeze” shift measured by LIGO-like detectors.
- **Scalar and Vector Memory:** In theories where the antisymmetric energy-momentum tensor is active, additional permanent shifts in the “breathing” (scalar) or “rotational” (vector) dimensions of the separation vector are possible.
- **BMS Symmetry:** Mathematically, the memory effect is deeply connected to the Bondi-Metzner-Sachs (BMS) group of symmetries at the boundary of spacetime. In GQFT, these symmetries must be satisfied for all five modes, suggesting that memory might be a universal feature of any unified gravitational field.

Detecting memory is exceptionally difficult because it manifests as a “DC” (zero-frequency) offset in the detector’s strain data. While individual events may have a memory signature too small for current LIGO sensitivity, researchers are developing “stacking” techniques—combining the data from hundreds of black hole mergers—to statistically identify the permanent displacement left behind by these spacetime ripples.

5. Advanced Mathematical Topics

5.1. The Stability of Minkowski Space. The linearized theory developed in Section 2.1 assumes that spacetime is “nearly” flat. However, to prove the

mathematical consistency of General Relativity, one must demonstrate the global stability of Minkowski space under the full, non-linear Einstein vacuum equations ($G_{\mu\nu} = 0$). This was famously proven by Christodoulou and Klainerman in 1993, providing a rigorous foundation for the existence of gravitational radiation.

Minkowski space $(\mathbb{R}^{1,3}, \eta)$ is the simplest solution to the vacuum Einstein equations. The stability problem asks whether “small” initial perturbations of the metric and its derivatives—representing incoming gravitational waves—evolve into a spacetime that remains close to Minkowski space for all future time.

- **Non-linearity:** Because gravity gravitates (waves carry energy that creates more curvature), there was a historical fear that even small waves might focus and form singularities (black holes).
- **Global Existence:** The proof required showing that solutions to the non-linear hyperbolic system of Einstein equations exist globally in time and do not “blow up”.

The core of the stability proof lies in the decay estimates of the curvature tensor. As gravitational waves propagate outward toward “null infinity” ($\mathcal{I}^+$), the curvature must dissipate at a specific rate.

- **Peeling Theorem:** In a stable Minkowski-like spacetime, different components of the Riemann tensor “peel off” at different rates as one moves away from the source.
- **Geometric Control:** Christodoulou and Klainerman used a framework of “optical functions” and “null frames” to track the energy flux of the waves, proving that the energy radiated to infinity is finite and that the manifold eventually relaxes back to a flat state.

Modern research has extended these results beyond the vacuum. For instance, the Einstein-Maxwell system is also globally stable, meaning that the interaction between gravitational waves and electromagnetic fields does not destabilize the background. This is particularly relevant for the GQFT framework, where the unified gravitational field contains additional scalar and vector components.

The stability of Minkowski space is what physically allows us to detect waves from the distant universe. If spacetime were unstable, the “chirps” from binary mergers would be distorted or absorbed by the background geometry before reaching Earth. Instead, the proof ensures that the waves we measure in Section 2.4 are true messengers of the distant, non-linear events that created them.

5.2. The Cauchy Problem and Initial Value Formulation. The Einstein Field Equations (EFE) are a set of ten coupled, non-linear partial differential equations. To solve them for a dynamic system—such as two black holes spiraling toward each other—we must formulate them as a Cauchy problem. This involves specifying the geometry of spacetime on a “snapshot” of time and then using the field equations to determine the geometry at all future moments.

The first step in this formulation is the 3+1 decomposition, developed by Arnowitt, Deser, and Misner (ADM). We “slice” the four-dimensional manifold $\mathcal{M}$ into a sequence of three-dimensional spacelike hypersurfaces Σ_t

- **The Induced Metric (g_{ij}):** This describes the intrinsic geometry of the 3D slice (the “snapshot”).
- **The Lapse Function (α) and Shift Vector (β^i):** These represent our freedom to choose how we move from one slice to the next—essentially how we “label” our time and space coordinates as we evolve the system.

When we break down the 4D Einstein equations into 3D components, they split into two distinct groups:

- **Constraint Equations:** These must be satisfied on every single slice. They ensure that the geometry we've chosen is physically consistent with the distribution of mass and energy (the Hamiltonian and Momentum constraints).
- **Evolution Equations:** These are the "equations of motion" for the metric. They dictate how the induced metric g_{ij} and its extrinsic curvature K_{ij} change as we move to the next time slice.

A critical mathematical requirement for the Cauchy problem is well-posedness: the guarantee that a small change in our initial "snapshot" results in only a small change in the future solution. For gravitational waves, this ensures that the ripples we predict are stable and unique. In the GQFT framework, the Cauchy problem is even more complex, as the initial data must also include the antisymmetric components of the gravitational field and their respective constraints to ensure the consistent evolution of all five propagating modes.

Just as we used the Lorenz gauge in the linearized theory, numerical solvers must choose a "gauge" (specific values for α and β^i) that prevents the coordinates from becoming singular. Choosing a "good" gauge is what allows a simulation to run all the way through a black hole merger without the math "crashing" before the peak of the gravitational wave signal is reached.

5.3. Gravitational Wave Memory and the Christodoulou Effect. The gravitational wave memory effect is a prediction that a burst of gravitational radiation leaves a permanent displacement in the relative positions of test masses. While the linear version of this effect was known earlier, the nonlinear memory effect—often called the Christodoulou effect—was a major discovery in the mathematical analysis of the Einstein equations.

To understand memory, we must look at the "boundary" of spacetime, specifically null infinity ($\mathcal{I}^+$). As gravitational waves propagate at the speed of light, they eventually reach this boundary.

- **The Bondi News:** This mathematical function describes the energy flux of the gravitational waves passing through null infinity.
- **Supertranslations:** The memory effect is deeply related to the BMS (Bondi-Metzner-Sachs) symmetry group. A gravitational wave essentially "shifts" the spacetime between two different supertranslation states, meaning the "flat" background before the wave is not exactly the same as the "flat" background after the wave.

In 1991, Demetrios Christodoulou proved that even if a source has no "linear" change (like a burst of particles), the gravitational waves themselves act as a source of gravity.

- **Waves Sourcing Waves:** Because the Einstein equations are nonlinear, the energy carried by the gravitational waves ($t_{\mu\nu}^{GW}$) creates its own curvature.
- **The "Nonlinear" Kick:** This self-interaction results in a permanent change in the metric perturbation Δh_{ij}^{TT} . In a binary black hole merger, the nonlinear memory actually provides the dominant contribution to the permanent displacement.

Recall the geodesic deviation equation from Section 2.4.1. If we integrate the acceleration $\ddot{S}^i$ twice over the duration of the event, the final separation S_{final}^i will differ from the initial $S_{initial}^i$:

$$\Delta S^i \approx \frac{1}{2} \Delta h_{ij}^{TT} S_{initial}^j.$$

This is not a “vibration” that fades; it is a permanent re-scaling of the distance between objects. If LIGO were sensitive enough to DC-offsets, the mirrors would remain at a slightly different distance from each other after every detected merger.

In Gravitational Quantum Field Theory (GQFT), the memory effect is expanded by the presence of the additional modes.

- **Scalar and Vector Memory:** Just as tensor waves carry energy to infinity, the scalar “breathing” and vector “spin” modes also contribute to the total energy flux at null infinity.
- **Unified Shift:** This implies that the total memory observed in a unified field theory would be a composite of tensor, vector, and scalar shifts, providing a potential experimental “smoking gun” for physics beyond General Relativity.

5.4. Numerical Relativity and High-Performance Computing. Numerical Relativity (NR) is the field of study that uses computer algorithms to solve the Einstein equations for highly dynamical, strong-field astrophysical systems. This was the final “missing link” in gravitational wave detection; without the high-precision waveforms generated by NR, LIGO would be unable to distinguish a true signal from background noise.

For decades, numerical simulations of binary black holes were unstable, usually “crashing” as the black holes approached one another. In 2005, Frans Pretorius achieved a breakthrough by using generalized harmonic coordinates and excision techniques. This allowed for the first full simulation of a binary merger—from the late inspiral, through the collision, to the final ringdown of a single black hole.

Solving the EFE on a computer requires sophisticated mathematical and engineering strategies:

- **Adaptive Mesh Refinement (AMR):** To save computing power, the simulation uses a coarse grid for the flat spacetime far from the source and a much finer, high-resolution grid near the black holes where the curvature is extreme.
- **Spectral Methods:** These represent the gravitational field as a sum of basis functions (like spherical harmonics). They are exceptionally efficient for the “ringdown” phase and are used extensively by the SXS (Simulating eXtreme Spacetimes) collaboration.
- **Parallelization:** These simulations are run on thousands of CPU and GPU cores simultaneously, as a single merger simulation can take weeks or even months to complete.

The primary output of NR is a waveform template bank.

- **Matched Filtering:** LIGO uses these templates to perform “matched filtering,” where the detector’s raw data is compared against thousands of pre-computed NR signals.
- **Parameter Estimation:** When a match is found, the specific template tells us the masses, spins, and distances of the objects involved.

In Gravitational Quantum Field Theory (GQFT), the numerical challenge is significantly amplified.

- **Extra Evolution Equations:** Instead of evolving just the symmetric metric, NR codes in a GQFT framework must evolve the antisymmetric field and the scalar potential alongside the standard tensor components.
- **Five-Mode Templates:** New templates are being developed to include the “breathing” (scalar) and “spin” (vector) signatures. If a signal deviates from the standard GR template, these multi-mode NR simulations will be the key to identifying the specific non-linear interactions predicted by unified field theories.

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